Partial Gluing and Level Two: Graded Explanations, Latent Actors and Paths through Singularities

Vinay K Awasthi
Johns Hopkins University
vawasth1@jh.edu

October 2026

Abstract

When local tables of evidence fail to glue into one global law, the usual response is to stop. We stay at the failure and ask how much structure can still be saved. For a family of tables with noncontextual fraction $\lambda^* < 1$, the laws that account for a fraction α of every table form a polytope E_α. It is non-empty exactly when $\alpha \le \lambda^*$, it shrinks as α grows, and its prediction intervals and its maximum-entropy member move continuously with α. The gradation α is how much of the gluing we keep; the rest, $1-\alpha$ of every table, is mass we let dissipate. We then go up one level, to laws over hypotheses: latent types of patients (actors) whose mixture produces the observed law. The explanations at this level are the real points of the fibre of a polynomial map, an affine scheme, and its geometry carries what happens to the actors. Where two types coincide the fibre is singular, a second-order obstruction κ (the first non-abelian term, $\ell_2 \neq 0$) is non-zero, and the tangent cone is a union of planes: types can split apart along one variable for free but not along two. Along a straight path of observed laws the types separate like $\sqrt{s}$ (two Puiseux branches); on the other side of the independence locus the two explanations become complex conjugates, no real two-type explanation exists, and a third type is needed. Going around a loop of populations can exchange the roles of the two actors ($\mathbb{Z}/2$ monodromy). With three measured variables this exchange must pass through a singularity; with four it need not. We illustrate everything on a drug-combination problem: baseline and single-drug harms are known, pairs have been trialled, the triple combination has not. Latent types narrow the prediction interval about fivefold; when the pair trials conflict, the graded explanations say what can be predicted when 20%, 40%, 70% or 86% of the structure is saved. Finally we make the scheme structure explicit. In mean-and-difference coordinates the model is a toric family over the weights; the fibre over an independent law is exactly the Stanley–Reisner scheme of a small simplicial complex whose faces are the strata (it is reduced); and for three variables the model inverts in closed form, flatness fails exactly where an actor is absent or the actors agree on a variable, and the sign of Cayley's hyperdeterminant separates genuine mixtures, signed mixtures and complex-conjugate pairs. Real explanations can vanish only at a wall of the parameter box or at the non-flat locus, and this is why role exchange needs a singularity with three variables but not with four. Both levels then sit on one ladder of base semirings, $\mathbb{C} \supset \mathbb{R} \supset \mathbb{R}_{\ge 0}$: disagreement on an overlap has no points at all; contextuality has real points but no positive ones, with negativity comparable to $1-\lambda^*$; and at level two a further rung of complex-conjugate actors appears. Finally we let the coefficients be matrices and keep what fails to commute: commutators decide the cover of contexts (CHSH gives $\lambda^* = 2-\sqrt{2}$), order effects are bounded by commutator norms, the canonical Kirkwood–Dirac section puts the commutation obstruction on the complex rung, loops carry a geometric phase that becomes non-abelian with rank-two projections, and non-commuting projections steer a state along a probability path (Zeno dragging) where commuting ones cannot.

With the measurements given but partly masked, the linear fit locates a hidden direction exactly when the known directions on its side span space, leaving otherwise a linear ambiguity of computable dimension; positivity then culls the candidates that would imply negative probabilities, and what remains recoverable are the invariants of the residual symmetry. Last, quantum graded gluing is the positivity domain of a hereditary quadratic matrix polynomial in the sense of Islam, Volčič and Woerdeman: its walls are determinantal free loci, and the quantum fraction lies below the classical fraction of every product-basis view, with equality for classical data and for pairwise Werner windows, where it exhibits monogamy. A final section gives these pieces a common categorical home: the contexts are a diagram of commutative subalgebras and contextuality a failed limit over it (Kochen–Specker versus Gleason, exact on the Peres–Mermin square); representation functors explain the gauge and show that a holonomy phase q allows finite dimension only when $q^n = 1$; positivity on truncated free modules (NPA) gives the quantum rung without fixing a dimension, with closed forms on isotropic boxes; and divergence-enriched categories give a data processing inequality for λ^*. All claims are checked: nine numerical checks (`engine/check_level2.py`) six exact symbolic ones (`engine/check_scheme.py`), six on the ladder (`engine/check_ladder.py`) eight on non-commutativity (`engine/check_noncomm.py`) seven on masked measurements (`engine/check_masked.py`, `engine/check_masked_gauge.py`) six on quantum graded gluing (`engine/check_qglue.py`) and six on the categorical home (`engine/check_npa.py`).

1 The problem: a combination nobody has trialled

Three drugs A, B, C are each known by themselves: single-drug trials give the rate of harm under each. Pairs have been trialled: from the AB, BC and AC trials we have the joint tables of harm. The triple combination has not been given to anyone. Write $H_A, H_B, H_C \in \{0, 1\}$ for whether a patient is susceptible to the harm of each drug, and suppose for the illustration that a patient on a combination is harmed when susceptible to at least one of its drugs. The question is

$$P(\text{any harm} \mid A{+}B{+}C) \;=\; 1 - P(H_A = H_B = H_C = 0),$$

a property of the joint law of (H_A, H_B, H_C), which no trial has observed. The baseline (no drug, no harm) and the single-drug margins are the known blobs; the joint law is the unknown blob; the pair tables are *windows* onto it.

Three things can happen.

1. The pair tables glue, and the set of global laws consistent with them gives an interval for the triple harm (level one).
2. Patients come in latent types, actors that are invisible in the data but shape it: a type susceptible to A and C, say, and a type susceptible to B. The same patient can play different roles in different pairs. If we are willing to assume a small number of types, the interval narrows (level two, section 3).
3. The pair tables do not glue: the BC trial was run in a different population. The standard answer is that there is no global law and nothing to predict. We stay at the failure and grade it (section 2).

The rest of the note develops each, and then follows the latent actors through the singular places of their geometry: where they merge, where one of them disappears, where a new one must be born, and where they exchange roles.

2 Sitting at the gluing failure: graded explanations

Let $X = (X_v)_{v \in V}$ take values in finite sets, and let a family of windows w have scopes $S_w \subseteq V$ and tables t_w, laws on the values of X_{S_w}. For a law q on the values of X write $q|_w$ for its marginal on S_w. The *noncontextual fraction* [3] is

$$\lambda^* \;=\; \max\{\, \mu(\text{all}) : \mu \geq 0 \text{ a measure on the values of } X,\ \mu|_w \leq t_w \text{ for every } w \,\},$$

a linear program. The tables glue (there is a global law with these marginals) iff $\lambda^* = 1$.

Definition 1 (graded explanation)**.** For $\alpha \in (0, 1]$, an *α-explanation* is a law q with $\alpha\, q|_w \leq t_w$ for every window. Write E_α for the set of them. For an event F, the *α-prediction interval* is $[\min_{q \in E_\alpha} q(F),\ \max_{q \in E_\alpha} q(F)]$ (two linear programs), and the *most probable explanation* q_α is the maximiser of the entropy $H(q)$ over E_α.

An α-explanation splits every table in two,

$$t_w \;=\; \alpha\, q|_w \;+\; (1-\alpha)\, r_w, \qquad r_w = \frac{t_w - \alpha q|_w}{1-\alpha} \ \text{ a law on } S_w, \tag{1}$$

a part $\alpha q|_w$ that comes from one global law, and a residue r_w that comes from no common source: the residues of different windows need not agree on their overlaps. The fraction α is the structure we save; $1 - \alpha$ of the mass of every table is let go. This is the gradation in gluing.

Proposition 2. *(i) $E_\alpha \neq \emptyset$ iff $\alpha \leq \lambda^*$.*

(ii) If $\alpha \leq \beta$ then $E_\beta \subseteq E_\alpha$. Hence prediction intervals are nested: saving more structure never widens them.

(iii) On $(0, \lambda^*]$ the map $\alpha \mapsto E_\alpha$ is continuous (in the Hausdorff metric), and so are the endpoints of every prediction interval and the most probable explanation q_α, which is unique.

(iv) If all table entries are positive, E_α is the whole simplex for small α: the interval is $[0, 1]$ for any event that is neither sure nor impossible, and q_α is the uniform law.

Proof. (i) If $q \in E_\alpha$ then $\mu = \alpha q$ is feasible for the program defining λ^* with total mass α, so $\alpha \leq \lambda^*$. Conversely, if μ attains λ^* and $\alpha \leq \lambda^*$, then $q = \mu/\lambda^*$ satisfies $\alpha q|_w \leq \lambda^* q|_w = \mu|_w \leq t_w$. (ii) $\alpha q|_w \leq \beta q|_w \leq t_w$. (iii) Let $M = \{\mu \geq 0 : \mu|_w \leq t_w\}$, a convex polytope; $E_\alpha = \{\mu/\alpha : \mu \in M,\ \mu(\text{all}) = \alpha\}$. The correspondence $\alpha \mapsto E_\alpha$ has a closed graph and compact values, so it is upper hemicontinuous. It is also lower hemicontinuous: given $q \in E_\alpha$ and $\alpha_n \to \alpha$, if $\alpha_n \leq \alpha$ then $q \in E_{\alpha_n}$ by (ii); if $\alpha < \alpha_n \leq \lambda^*$, take $\mu^* \in M$ of mass λ^* and $s_n \to 0$ with $(1 - s_n)\alpha + s_n \lambda^* = \alpha_n$; then $\mu_n = (1 - s_n)\alpha q + s_n \mu^* \in M$ (by convexity) has mass α_n, and $q_n = \mu_n/\alpha_n \in E_{\alpha_n}$ tends to q. A continuous compact-valued correspondence has continuous maxima and minima of a continuous function (Berge's maximum theorem [5]); the entropy is strictly concave, so its maximiser is unique and continuous. (iv) If every entry of every table is positive, every law q satisfies $\alpha q|_w \leq t_w$ once $\alpha \leq \min t_w$, so E_α is the whole simplex for small α; zero entries in the tables force the corresponding cells to zero for every α, and the statement holds on the remaining face. □

So the graded explanations form a path. At $\alpha \to 0$ we have saved nothing and know nothing; at $\alpha = \lambda^*$ we have saved all the structure that can be saved together. In between, $\alpha \mapsto q_\alpha$ is the *most probable path* from ignorance to the best partial gluing, and the prediction band around it narrows monotonically.

Example 3 (conflicting pair trials)**.** Let the truth in population 1 be two patient types with weights $(0.6, 0.4)$ and susceptibilities

$$a_1 = (0.5,\ 0.1,\ 0.4), \qquad a_2 = (0.05,\ 0.5,\ 0.1) \qquad (\text{for } H_A, H_B, H_C),$$

the harms independent within a type. The AB and AC trials sample population 1; the BC trial samples population 2, with the same types in proportions $(0.25, 0.75)$. Each table is exact, but together they do not glue: $\lambda^* = 0.86$. The triple-combination harm is 0.667 in population 1 and 0.612 in population 2. Table 1 and fig. 1(a) show the graded answer.

structure saved α	prediction interval	most probable q_α	reading
0.2	[0, 1]	0.867	nothing ruled out; close to the uniform law (7/8)
0.4	[0, 1]	0.835	
0.6	[0.233, 0.976]	0.781	the trials begin to bite
0.7	[0.343, 0.836]	0.765	
0.8	[0.488, 0.732]	0.694	
$0.86 = \lambda^*$	[0.594, 0.681]	0.647	all that can be saved; both populations' truths inside
0.87	none	none	cannot be saved together

Table 1: Graded partial gluing of the conflicting pair trials (example 3): the triple-combination harm when a fraction α of every table is explained by one global law and $1 - \alpha$ is let go.

At $\alpha = \lambda^*$ the interval $[0.594, 0.681]$ contains the triple harm of both populations. This is not a theorem (it is a property of this example), but it shows what the graded answer is for: when the evidence conflicts, the interval at the largest savable fraction is an honest statement of what is still predictable, and the residues r_w in (1) say where the conflict lives.

3 Level two: latent actors

Level one asks for a law on outcomes. Level two asks for a law on hypotheses that produces it: here, a mixture of patient types.

Definition 4 (latent class model)**.** With k types and m binary variables, type h has weight w_h and susceptibilities $a_h \in [0,1]^m$, the variables independent given the type. The observed law is

$$F(w, a) \;=\; \sum_{h=1}^{k} w_h\, \Pi(a_h), \qquad \Pi(a_h) = \mathrm{Bern}(a_{h1}) \otimes \cdots \otimes \mathrm{Bern}(a_{hm}),$$

a polynomial map from $\Delta_{k-1} \times [0,1]^{km}$ to the simplex of laws on $\{0,1\}^m$.

For an observed law P, the level-two explanations are the real points, with $w \geq 0$ and $a \in [0,1]^{km}$, of

$$\mathcal{X}_P \;=\; \mathrm{Spec}\ \mathbb{R}[w, a] \big/ \big(\textstyle\sum_h w_h - 1,\ F(w, a) - P \big),$$

the fibre of F over P, an affine scheme. When only windows are observed, F is composed with the marginal maps, and the fibre is taken over the observed tables. Level one is the image; level two is the fibre. Everything that happens to the actors happens in the fibre: a fibre that is a finite

set of points (identification), a curve (a one-parameter family of explanations), a point where two branches cross (actors that merge), or empty (no explanation with this many actors).

For $k = 2$ and $m = 3$ there are $1 + 2m = 7$ parameters and the laws on $\{0,1\}^3$ form a 7-dimensional simplex; generically the fibre over the full law is two points, exchanged by swapping the labels of the types [4]. Over the three pair tables (6 numbers after normalisation) the fibre is a curve.

Example 5 (latent structure sharpens prediction)**.** In population 1 of example 3, with all three pair trials run there, the pair tables glue. Level one (any global law with these pair tables) gives the triple harm in [0.654, 0.694]. Level two (any mixture of two types with these pair tables) gives [0.6613, 0.6687], about five times narrower, and still containing the truth 0.667 (fig. 1(b)). The level-two interval was computed by tracing the fibre curve: we fix w_1 on a grid of 49 values and solve for the susceptibilities from 12 starts each; 228 exact solutions were found.

Assuming latent actors is a modelling choice and the narrower interval is only as good as that choice. What the fibre adds is that the choice is explicit and its consequences can be computed, including where it breaks.

Graded gluing at level two. The gradation of section 2 lifts: the level-two gluable fraction is

$$\lambda_2^* \;=\; \sup\{\alpha :\; \alpha\, F(w,a)|_w \le t_w \text{ for every window, for some } (w,a)\},$$

the largest share of every table that a k-type population can account for. Since $F(w,a)$ is a law, $\lambda_2^* \le \lambda^*$. For the conflicting trials of example 3 the best two-type explanation found saves 0.86, the same as level one: there, assuming two actors costs no structure. Unlike λ^*, λ_2^* is not a linear program (it is a polynomial optimisation), and the values we report are the best found from many starts, so they are lower bounds.

4 Singularities and the second-order obstruction

Let $J = dF$ at a point $x = (w,a)$, restricted to directions with $\sum_h \delta w_h = 0$, and viewed as a map into the sum-zero directions of the simplex. Its kernel K is the space of first-order deformations of the explanation that leave the observed law unchanged; let C be a basis of its cokernel. Define the *second-order obstruction*

$$\kappa(v) \;=\; C^\top D^2F(v,v), \qquad v \in K.$$

Proposition 6 (Kuranishi test [19])**.** *If $\gamma(t)$ is a smooth path in the fibre with $\gamma(0) = x$ and $\gamma'(0) = v$, then $v \in K$ and $\kappa(v) = 0$.*

Proof. Differentiate $F(\gamma(t)) = P$ twice at $t = 0$: $Jv = 0$ and $J\gamma''(0) + D^2F(v,v) = 0$, so $D^2F(v,v) \in \operatorname{im} J$, which $C^\top$ annihilates. □

At a regular point $K = 0$ and nothing is obstructed. The obstruction is the first term of the deformation problem that is not linear: in the language of L_∞ algebras [20] it is the bracket ℓ_2, and at level one (where the explanation sets are polytopes) it vanishes identically. Level two is where it first appears. The fibre has three kinds of points (table 2).

point	meaning	rank J	dim K	obstruction
generic	two distinct, present actors	$2m+1=7$	0	none
$w_2=0$	an actor absent	$m+1=4$	$m=3$	$\kappa=0$: its role is free
$a_1=a_2$	two actors coincide	$m=3$	$m+1=4$	$\kappa\neq 0$ off a union of planes

Table 2: Strata of the two-type, three-variable fibre ($m=3$). An absent actor can take any role at no cost; coinciding actors can be pulled apart only along special directions.

Theorem 7 (tangent cone at a coincidence)**.** *Let $a_1=a_2=c\in(0,1)^m$ and $w\in(0,1)^2$. The kernel K is spanned by the weight direction $(\delta w,0)$ with $\delta w=(1,-1)$ and the* splits $s(e)=(0;\ -\frac{w_2}{w_1}e,\ e)$, $e\in\mathbb{R}^m$. *For $v=t\,(\delta w,0)+s(e)$,*

$$\kappa(v)\ =\ \frac{2w_2}{w_1}\sum_{i<j}e_ie_j\ C^\top\partial_{ij}\Pi(c),$$

which vanishes iff e has at most one non-zero coordinate. The tangent cone of the fibre at the coincidence is therefore the union of the m planes $\langle(\delta w,0),\ s(\mathbf{e}_i)\rangle$, which meet along the weight line, and each plane is filled by exact paths in the fibre.

Proof. Write $\partial_S\Pi(c)$ for Π with the factors $i\in S$ replaced by $(-1,1)$. The 2^m vectors $\partial_S\Pi(c)$ form a basis (a tensor product of bases $\{\mathrm{Bern}(c_i),(-1,1)\}$), and those with $S\neq\emptyset$ span the sum-zero directions. The columns of J are $\Pi(a_1)-\Pi(a_2)=0$ (weights) and $w_h\partial_i\Pi(c)$ (susceptibilities), so $\operatorname{im}J=\langle\partial_i\Pi\rangle$, the kernel is as stated, and the cokernel is spanned by $\partial_S\Pi$ with $|S|\geq 2$. In $D^2F(v,v)=2\sum_h\sum_i\delta w_h\delta a_{hi}\partial_i\Pi+2\sum_h w_h\sum_{i<j}\delta a_{hi}\delta a_{hj}\partial_{ij}\Pi$ (there are no ∂_{ii} terms, F being multilinear in each a_h), the first sum lies in $\operatorname{im}J$ and the second equals $2\big(w_1\frac{w_2^2}{w_1^2}+w_2\big)\sum_{i<j}e_ie_j\partial_{ij}\Pi=\frac{2w_2}{w_1}\sum_{i<j}e_ie_j\partial_{ij}\Pi$. Since the $\partial_{ij}\Pi$ are independent modulo $\operatorname{im}J$, $\kappa(v)=0$ iff $e_ie_j=0$ for all $i<j$. For the paths: if two types differ only in variable i, the mixture is still a product law, whose i-th margin is $w_1a_{1i}+w_2a_{2i}$; moving the weights and spreading a_{1i},a_{2i} while keeping this mean fixed stays in the fibre exactly. □

A mixture of two products that differ in one variable is itself a product. Actors that differ in only one respect are therefore invisible: the fibre contains a whole plane of them through the coincidence. Actors that differ in two respects or more create a correlation, and they cannot be separated infinitesimally without changing the observed law at second order. Figure 1(d) shows $|\kappa|$ around the circle of splits in the (X_1,X_2) plane: it vanishes exactly on the two coordinate axes and is proportional to $|\sin 2\varphi|$ in between.

5 Where mass dissipates: paths through singularities

Now move the observed law. Fix $w=(\frac12,\frac12)$, $c\in(0,1)^3$ and $e\in\mathbb{R}^3$ with all $e_i\neq 0$, and set $B=\sum_{i<j}e_ie_j\,\partial_{ij}\Pi(c)$ and $P(s)=\Pi(c)+sB$, a straight line of laws through the product law $\Pi(c)$.

Theorem 8 (two branches, then none)**.** *(i) For $s>0$ (small enough that the parameters stay in $[0,1]$), $P(s)=F\big(w,\ (c+\sqrt{s}\,e,\ c-\sqrt{s}\,e)\big)$ exactly. The two types separate like $\sqrt{s}$, and the two Puiseux branches $\pm\sqrt{s}$ [13] are exchanged by swapping labels.*

(ii) For $s<0$ no mixture of two (or one) product laws equals $P(s)$.

(iii) At $s=0$ the path passes through the coincidence stratum of theorem 7.

Proof. (i) $\Pi(c+te) = \sum_S t^{|S|} e_S\, \partial_S \Pi(c)$ with $e_S = \prod_{i\in S} e_i$. The average of t and $-t$ keeps the even terms: $\Pi(c) + t^2 B$ for $m = 3$. Put $t^2 = s$. (ii) Under $P(s)$ the covariance of X_i, X_j is $s\, e_i e_j$, so the product of the three pairwise covariances is $s^3 (e_1 e_2 e_3)^2 < 0$. Under any two-type law it is $\prod_{i<j} w_1 w_2 d_i d_j = (w_1 w_2)^3 (d_1 d_2 d_3)^2 \geq 0$, where $d = a_1 - a_2$. □

This is the precise form of an abrupt change. The observed law moves linearly in s; the explanation moves like $\sqrt{s}$, with infinite speed at $s = 0$; and on the far side the two-type structure is gone from the real world: the same formula gives $c \pm i\sqrt{|s|}\, e$, a complex-conjugate pair of explanations (section 7 makes this exact). In the space of observed laws the image of F has a fold along the independence locus, and the path crosses it. Figure 1(c) shows the gap $\max_i |a_{1i} - a_{2i}| = 2\sqrt{s}\max_i |e_i|$ fitted from the data, exactly on the curve, and the best two-type misfit for $s < 0$: 0.0004, 0.0008, 0.0017 and 0.0035 at $s = -0.005, -0.01, -0.02, -0.04$.

Graded gluing says how much is lost. Beyond the fold $\lambda_2^* < 1$: some mass of every observed law must be let go if two actors are to explain the rest. The best two-type explanations we found save 0.9990, 0.9926, 0.9960, 0.9919 and 0.9836 of the structure at $s = -0.0025$, -0.005, -0.01, -0.02, -0.04: about 1% of the mass must go, a little more as s moves away from the fold. These are attained values and so lower bounds on λ_2^*; that they are not monotone in s shows the optimiser does not resolve the exact rate. The mass does not vanish into nothing, however: three types fit $P(s)$ exactly for $s = -0.01$ and $s = -0.04$ (residual below 10^{-14}). What dissipates is the two-actor organisation of the mass, and what is born on the far side is a third actor. Read in the other direction, the fold is where a type dies: as $s \downarrow 0$ the two types merge, and their difference, which had carried a correlation, becomes a free direction of the kind in table 2.

6 Dual roles and monodromy

The labels "type 1" and "type 2" are not in the data. Over a generic law the fibre is two points that differ only in the labels, and along a path of laws the labels can be carried by continuity. Around a closed loop of laws they may come back exchanged: the actor that played the A-susceptible role now plays the B-susceptible one. This is the monodromy of the map F, here a subgroup of $\mathbb{Z}/2$ (of the symmetric group S_k for k types).

We build such loops explicitly. With a centre c, directions $u, v \in \mathbb{R}^m$, $r = 0.22$ and $\rho = 0.15$, put

$$a_{1,2}(\theta) = c \pm r(\cos\theta\, u + \sin\theta\, v), \qquad w(\theta) = \big(\tfrac{1}{2} + \rho\cos\theta,\ \tfrac{1}{2} - \rho\cos\theta\big), \qquad \theta \in [0, \pi].$$

At $\theta = \pi$ the parameters are those at $\theta = 0$ with the labels swapped, so the observed law $F(\theta)$ returns to its start (to 10^{-12}): a half circle of parameters is a closed loop of observed laws, and tracking the explanation along it exchanges the actors.

Theorem 9 (three variables force a singularity)**.** *Let $m = 3$, $k = 2$. Any continuous path of parameters with $w \in (0,1)^2$ from (w, a_1, a_2), with all $d_i = a_{1i} - a_{2i} \neq 0$, to its relabelling (w_2, w_1, a_2, a_1) passes, for each variable i, through a point where $d_i = 0$, and there* $\operatorname{rank} J \leq 5 < 7$. *So within the regular locus no role exchange is possible.*

Proof. Along the path d_i goes from d_i to $-d_i$ and so vanishes somewhere. Where $d_3 = 0$, say, the law is $G(w, a_{\cdot,1:2}) \otimes \mathrm{Bern}(c_3)$, where G is the two-type model on the first two variables. G has 5 parameters and a 3-dimensional target, so its differential has a kernel of dimension at least 2; those directions (with $\delta a_{\cdot 3} = 0$) are in the kernel of J. □

With four variables the same argument leaves a two-type model on three variables, which is generically regular, and nothing forces a singularity. Our 4-variable loop ($u = (1, 0.6, -0.4, 0.2)$, $v = (0.3, 1, 0.8, -0.9)$, $c = \frac{1}{2}\mathbf{1}$) keeps the smallest singular value of J at or above 0.0072 along 500 points of the half loop, the tracked solution ends at the swapped labels, and the full loop returns the original labels: genuine $\mathbb{Z}/2$ monodromy in the regular locus (fig. 1(e,f)). This is numerical evidence for the loop, not a proof. The 3-variable loop ($u = (1, 0.6, -0.4)$, $v = (0.3, 1, 0.8)$) shows the three forced dips of theorem 9, one for each variable, where the smallest singular value falls to the order of 10^{-6}.

In the drug setting: with three measured harms, two patient types cannot exchange their roles without passing through a population in which they look alike in some harm; with four harms they can. Which actor is "the one susceptible to A" is a local name, defined along a path, not a global fact; the observed law, at level one, is single-valued and never notices.

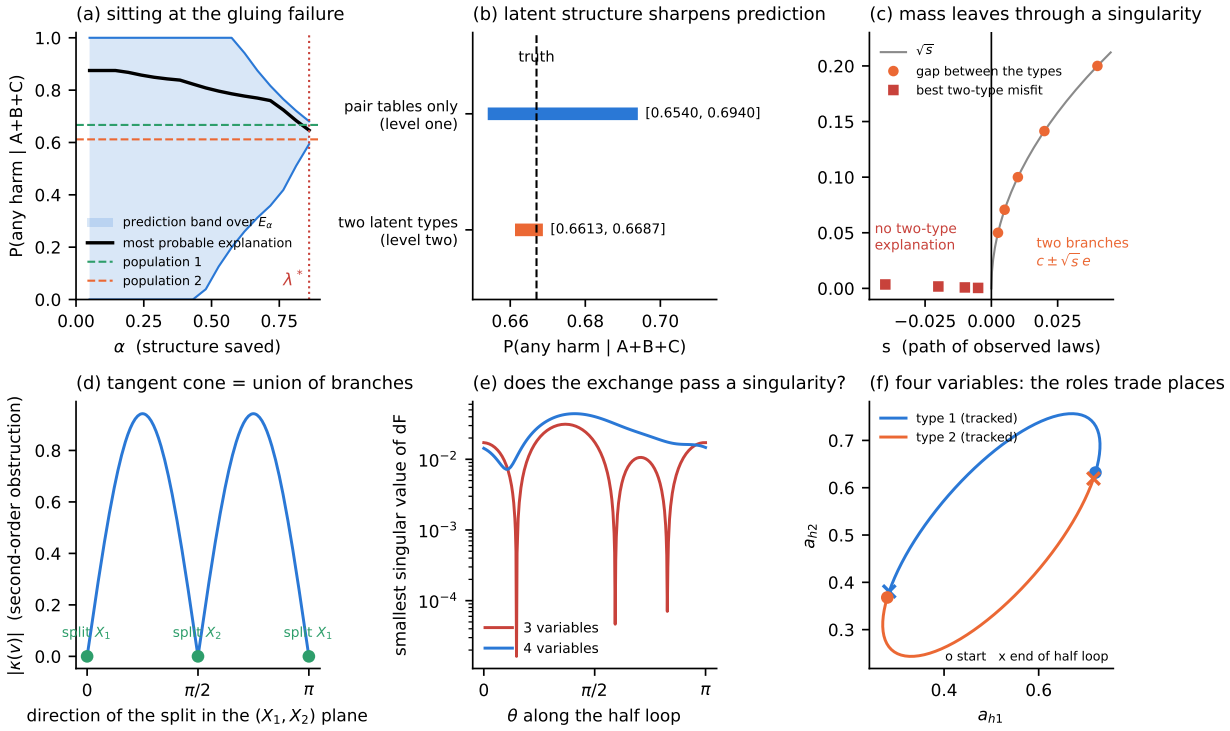


Figure 1: (a) Graded explanations for the conflicting trials: the prediction band over E_α, the most probable explanation, and the two populations' truths. (b) Level one against two latent types (population 1). (c) Crossing the independence locus: the gap between the types follows $\sqrt{s}$; for $s < 0$ no two-type explanation fits. (d) The second-order obstruction around the circle of splits at a coincidence: zero exactly on the coordinate branches. (e) Smallest singular value of dF along a role-exchanging half loop: with three variables it dips to zero three times, with four it stays away from zero. (f) The 4-variable half loop: the tracked types trade places.

7 Scheme structure: the fibre over independence and the flattening stratification

So far "scheme" has named the equations of the fibre. This section uses the scheme structure: ideals, their decompositions, reducedness and flatness. All statements here are checked exactly by Gröbner-basis computations [15] in SymPy [29] (`engine/check_scheme.py`, SC1–SC6).

Coordinates. Replace the susceptibilities by the mean and the difference of the two types,

$$\mu = w_1 a_1 + w_2 a_2, \qquad d = a_1 - a_2, \qquad \text{so} \qquad a_1 = \mu + w_2 d, \quad a_2 = \mu - w_1 d,$$

a polynomial automorphism of the parameter space (given w). On the observed side use the mean μ and the central moments $m_S = E\prod_{i\in S}(X_i - \mu_i)$, $|S| \geq 2$: the law and (μ, m) determine each other polynomially (unitriangularly).

Proposition 10 (moment normal form; SC1)**.** *$m_S = c_{|S|}(w)\, d_S$ with $d_S = \prod_{i\in S} d_i$ and $c_k(w) = w_1 w_2^k + w_2(-w_1)^k = w_1 w_2\big(w_2^{k-1} - (-w_1)^{k-1}\big)$.*

Proof. $a_{1i} - \mu_i = w_2 d_i$ and $a_{2i} - \mu_i = -w_1 d_i$, so $m_S = w_1(w_2)^{|S|} d_S + w_2(-w_1)^{|S|} d_S$. □

For fixed weights the map $d \mapsto (c_{|S|} d_S)_S$ is monomial: the two-type model is a toric family over the line of weights. These are the *secant cumulants* of Michałek, Oeding and Zwiernik [24], in which the secant variety becomes toric. Note $c_2 = w_1 w_2$ and $c_3 = w_1 w_2 (w_2 - w_1)$: an evenly mixed pair leaves no three-way moment.

The fibre over an independent law. Let $P = \Pi(c)$ be a product law (the variables independent). Its fibre is where the actors have merged, or one has gone.

Theorem 11 (the fibre is a Stanley–Reisner scheme; SC2)**.** *The ideal of the fibre over $\Pi(c)$ is*

$$I = (\mu - c) \;+\; \big(w_1 w_2\, d_i d_j :\; i < j\big) \;=\; (\mu - c) + \Big[(w_1) \cap (w_2) \cap \bigcap_{i=1}^{m} (d_j :\; j \neq i)\Big].$$

It is an intersection of prime ideals, hence radical: the fibre scheme is reduced. With $\mu = c$ it is the Stanley–Reisner scheme (cut by $w_1 + w_2 = 1$) of the simplicial complex Δ on the vertices $\{w_1, w_2, d_1, \ldots, d_m\}$ with facets

$$\{w_1, d_1, \ldots, d_m\}, \qquad \{w_2, d_1, \ldots, d_m\}, \qquad \{w_1, w_2, d_i\}\ (i = 1, \ldots, m).$$

Proof. By proposition 10 the fibre is $\mu = c$ and $c_{|S|}(w) d_S = 0$ for all $|S| \geq 2$. Every c_k is divisible by $c_2 = w_1 w_2$ and every d_S with $|S| \geq 3$ by some $d_i d_j$, so the ideal is $(\mu - c) + (w_1 w_2 d_i d_j)$. Since w_1 and $w_2 = 1 - w_1$ are coprime, $(w_1 w_2) = (w_1) \cap (w_2)$; the ideal $(d_i d_j : i < j)$ is the Stanley–Reisner ideal of m isolated points, $\bigcap_i (d_j : j \neq i)$; and the two involve disjoint variables, so their intersection is their product. The facets are the complements of the minimal non-faces $\{w_1, w_2, d_i, d_j\}$ in the sense of Stanley–Reisner theory [25]. The computation was also checked directly (Gröbner bases for $m = 3, 4$, and the intersection by elimination). □

The facets are the strata of table 2 named combinatorially: $\{w_1, d_1, \ldots, d_m\}$ is "actor 2 absent, actor 1's difference free", $\{w_2, \ldots\}$ the reverse, and $\{w_1, w_2, d_i\}$ "both present, differing only in variable i". Away from $w_1 w_2 = 0$ the factor $w_1 w_2$ is a unit, so near an interior coincidence the fibre

is *exactly* the Stanley–Reisner scheme of m points times the weight line. This upgrades theorem 7 from the tangent cone to the fibre itself, and since the scheme is reduced there are no nilpotent directions that the second-order test κ could have missed. The simplicial complexes of the gluability filtration in the companion report [31] and the singular fibres of level two are therefore the same kind of object.

Three variables: closed form, flatness and the hyperdeterminant. For $m = 3$ the map $(w_1, \mu, d) \mapsto (\mu, m_{12}, m_{13}, m_{23}, m_{123})$ goes between spaces of dimension 7 and can be inverted. Put $\Delta_m = m_{12}m_{13}m_{23}$ and $\mathrm{Det} = m_{123}^2 + 4\Delta_m$.

Proposition 12 (SC3, SC4)**.** *(i) With* $\{i, j, k\} = \{1, 2, 3\}$,

$$w_1 w_2 = \frac{\Delta_m}{\mathrm{Det}}, \qquad w_2 - w_1 = \pm\sqrt{1 - \frac{4\Delta_m}{\mathrm{Det}}}, \qquad d_i = \frac{m_{123}}{(w_2 - w_1)\, m_{jk}};$$

the sign of the square root is the labelling.

(ii) The Jacobian determinant is $-(w_1 w_2)^3 (d_1 d_2 d_3)^2$.

(iii) Det *is Cayley's hyperdeterminant [17] of the* $2 \times 2 \times 2$ *table of the law.*

Theorem 13 (flattening stratification, $m = 3$; SC5)**.** *Over the space of laws:*

laws	*fibre*	*flat*
all $m_{ij} \neq 0$, $\mathrm{Det} \neq 0$	*two reduced points (labels swap)*	*yes (étale, degree 2)*
one variable independent of the other two	*a surface*	*no*
product laws	*Stanley–Reisner scheme (dim. 3)*	*no*
exactly one $m_{ij} = 0$*; or* $\mathrm{Det} = 0 \neq \Delta_m$	*empty (closure only)*	—

On the parameter side the map is flat exactly off $\{w_1 w_2 d_1 d_2 d_3 = 0\}$*: where an actor is absent or the actors agree on a variable. On the generic stratum the real points are: two genuine mixtures* ($w \in (0, 1)$) *if* $\Delta_m > 0$*; two signed mixtures (real, but* $w \notin [0, 1]$*) if* $\mathrm{Det} > 0 > \Delta_m$*; a complex-conjugate pair if* $\mathrm{Det} < 0$.

Proof. The fibres are read off proposition 12 and checked by Gröbner bases. Flatness: source and target are regular of the same dimension, so by "miracle flatness" [23, Thm. 23.1] the map is flat at a point iff the fibre through it has dimension 0 there; the Jacobian vanishes exactly on $w_1 w_2 d_1 d_2 d_3 = 0$, and those points lie in the positive-dimensional fibres of the second and third rows. Real points: $p = w_1 w_2 = \Delta_m/\mathrm{Det}$ is real and $1 - 4p = m_{123}^2/\mathrm{Det}$. If $m_{123} \neq 0$, w is real iff $\mathrm{Det} > 0$, and then d is real by (i). If $m_{123} = 0$, then $w = (\frac{1}{2}, \frac{1}{2})$ and $d_i^2 = m_{ij}m_{ik}/(p\, m_{jk})$ has the sign of Δ_m, which here is the sign of $\mathrm{Det} = 4\Delta_m$. In both cases $w \in (0, 1)$ iff $p \in (0, \frac{1}{4}]$ iff $\Delta_m > 0$. The sign of Det separates real tensor rank 2 from 3, as in de Silva and Lim [16]. □

This corrects the picture of section 5. The fold where real explanations disappear is *not* where flatness fails: over $\mathbb{C}$ the two explanations persist and become conjugate. What the stratification gives is the following.

Corollary 14 (where explanations can die)**.** *Along a continuous path of laws, a two-type probability explanation (parameters in the box* $w \in [0, 1]$*,* $a \in [0, 1]^{2m}$*,* $m = 3$*) can cease to exist only (a) by reaching a wall of the box, or (b) at a law where its limit lies in the non-flat locus* $\{w_1 w_2 d_1 d_2 d_3 = 0\}$.

Proof. On $\{\Delta_m > 0\}$ the inverse is continuous (proposition 12, $\mathrm{Det} \geq 4\Delta_m > 0$), so an explanation persists while the path stays there and inside the box. If the path leaves $\{\Delta_m > 0\}$ at a law P^* while the explanations stay in the compact box, a limit point x^* has $F(x^*) = P^*$ with $\Delta_m(P^*) = (w_1 w_2)^3 (d_1 d_2 d_3)^2 = 0$. □

The box wall is the semialgebraic part, invisible to the scheme; the inequalities that cut the probability image out of the secant variety are those of Allman, Rhodes, Sturmfels and Zwiernik [12] (see also [26]).

Corollary 15 (monodromy from the stratification)**.** *Over $\{\Delta_m > 0\}$ the sign of d_1 is a continuous, nowhere-zero label of the two sheets, so the real double cover of genuine mixtures is trivial: roles cannot be exchanged without leaving through the non-flat locus (theorem 9). Over $\mathbb{C}$ the cover is connected and its monodromy is $\mathbb{Z}/2$. For $m = 4$, a variable independent of the rest ($d_4 = 0$) is a flat point (full rank, two points; SC6), the non-flat walls move to codimension ≥ 2 and no longer separate the real region, which is why the 4-variable loop of section 6 can exchange roles.*

We expect that for every $m \geq 3$ the map is flat exactly where $w_1 w_2 \neq 0$ and at least three d_i are non-zero (consistent with [4]); we have checked this only for $m = 3$ and at a point for $m = 4$.

The image. The Zariski closure of the image of the k-type model is the k-th secant variety of the Segre embedding of $(\mathbb{P}^1)^m$ [21]. For $k = 2$, $m = 3$ it fills $\mathbb{P}^7$ (generic $2 \times 2 \times 2$ tensors have rank 2), which is why the boundary of the two-type region is real and semialgebraic (the walls $\Delta_m = 0$, $\mathrm{Det} = 0$ and the box), not Zariski-closed. For $m \geq 4$ the secant variety has dimension $2m + 1 < 2^m - 1$ and is a proper subvariety, toric in secant cumulants [24]. Figure 2 shows the three real regions on a slice, the complex Δ, and a path crossing the walls.

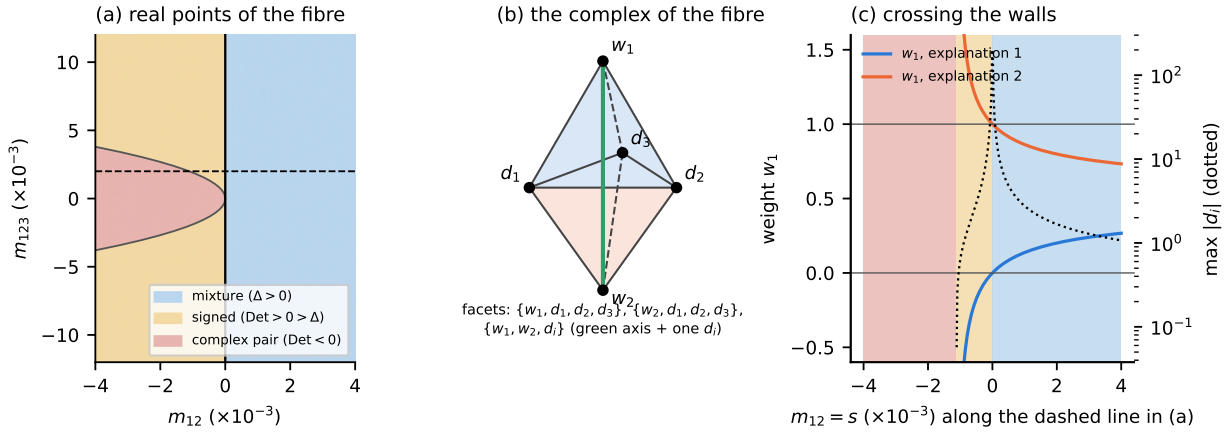


Figure 2: (a) A slice of the law space ($m_{13} = m_{23} = 0.03$, mean fixed) coloured by the real points of the two-type fibre: genuine mixtures ($\Delta_m > 0$), signed mixtures ($\mathrm{Det} > 0 > \Delta_m$), complex-conjugate pairs ($\mathrm{Det} < 0$, inside the parabola $m_{123}^2 = -4\Delta_m$). (b) The simplicial complex Δ of theorem 11 for $m = 3$: two tetrahedra on the triangle $d_1 d_2 d_3$, and the axis $w_1 w_2$ with the triangles $\{w_1, w_2, d_i\}$; its Stanley–Reisner scheme is the fibre over an independent law. (c) Along the dashed line of (a): the weights of the two explanations (they leave $[0, 1]$ as the path crosses $m_{12} = 0$, where one weight tends to 0 and $\max |d_i|$ blows up, and they are complex beyond $\mathrm{Det} = 0$). A probability explanation also needs $|d_i| \leq 1$, so in practice it hits the box wall before $m_{12} = 0$.

8 The ladder of base semirings

The two levels fail in the same way: their equations have points over a larger number system but not over the non-negative one. We make this a single picture. Write the evidence at either level as polynomial equations with rational coefficients, and present a probability a together with its complement $\bar{a} = 1 - a$, so that the parameter box is cut out by $a, \bar{a} \geq 0$. Then ask over which semiring the equations have points:

$$\mathbb{C} \supset \mathbb{R} \supset \mathbb{R}_{\geq 0}.$$

An $\mathbb{R}_{\geq 0}$-point is a probability explanation; an $\mathbb{R}$-point that is not one is a signed explanation (a quasi-probability, or actors with negative weight); a $\mathbb{C}$-point that is not real is an oscillatory explanation; and no point at all is the bottom of the ladder. Each rung has its own measure of how badly the next rung up fails (table 3).

rung	level one (a law on outcomes)	level two (two latent actors, three variables)
none	the tables disagree on an overlap (signalling, a direct influence); measured by the *linear gap*, the least ℓ_1 repair	not in the image: one pairwise moment zero alone, or $\mathrm{Det} = 0 \neq \Delta_m$
$\mathbb{C}$	same as $\mathbb{R}$ (linear equations)	$\mathrm{Det} < 0$: a complex-conjugate pair; measured by the imaginary part
$\mathbb{R}$	contextual: a signed global section exists, no positive one; measured by the *negativity* ν	signed mixture, $\mathrm{Det} > 0 > \Delta_m$; measured by the negative weight $\nu_2 = \frac{1}{2}\left(\lvert m_{123}\rvert/\sqrt{\mathrm{Det}} - 1\right)$
$\mathbb{R}_{\geq 0}$	the tables glue ($\lambda^* = 1$)	genuine mixture, $\Delta_m > 0$; then the box $a \in [0,1]$ is a further semialgebraic wall

Table 3: The ladder at both levels. Level one has no complex rung, because its equations are linear; level two has one, because they are not.

Level one. Let $A\mu = t$ be the marginal equations of a family of windows, with μ a signed measure of total mass 1. The *linear gap* is $\min \|A\mu - t\|_1$, and when it is zero the *negativity* is $\nu = \min\{\|\mu^-\| : A\mu = t\}$, the least total negative mass of a signed global section; both are linear programs. The negativity is the quantity behind Abramsky and Brandenburger's operational reading of negative probabilities [2].

Theorem 16 (the ladder at level one; LD1–LD3)**.** *(i) $\mathbb{C}$-points exist iff $\mathbb{R}$-points exist iff the linear gap is 0; for the measurement scenarios we test (cycles, CHSH, three outcomes) this holds whenever the tables agree on their overlaps [1].*

(ii) $\mathbb{R}_{\geq 0}$-points exist iff $\nu = 0$ iff $\lambda^* = 1$.

(iii) On the set of families with linear gap 0 (a polytope), ν and $1 - \lambda^*$ are comparable: there are constants $0 < c_- \leq c_+$ depending only on the scenario with $c_-(1 - \lambda^*) \leq \nu \leq c_+(1 - \lambda^*)$. One can take $c_+ = \nu(r)$ maximised over the no-signalling polytope.

Proof. (i) The equations are linear with rational coefficients, so solvability over $\mathbb{C}$, $\mathbb{R}$ and $\mathbb{Q}$ coincide; the gap is the ℓ_1 distance to solvability. (ii) A signed section with $\nu = 0$ is a global law, and conversely. (iii) Upper bound: write $t_w = \lambda^* q|_w + (1 - \lambda^*) r_w$ as in (1). The residues r_w agree on overlaps because t and $q|_w$ do, so r has a signed section μ_r, and $\lambda^* q + (1 - \lambda^*)\mu_r$ is a signed section of t with negativity $(1 - \lambda^*)\nu(\mu_r)$. Lower bound: ν and $1 - \lambda^*$ are both convex and piecewise linear in t (values of linear programs with t in the right-hand side), and they vanish on the same polytope,

the noncontextual polytope. A polyhedral convex function grows at least linearly away from its zero set (weak sharp minima [6], a form of Hoffman's bound [28]), and the other is Lipschitz and vanishes there; so each is bounded by a multiple of the distance to that polytope, and hence by a multiple of the other. □

The constants depend on the scenario, and we measured them (fig. 3(a), 30 random mixtures of a global law with a contextual family in each of five scenarios). The ratio $\nu/(1-\lambda^*)$ is exactly $\frac{1}{2}$ for the 3-cycle, the 4-cycle and CHSH [14] with the Popescu–Rohrlich box [10], exactly $\frac{1}{4}$ for the anti-correlated 5-cycle, and between $\frac{1}{2}$ and $\frac{2}{3}$ for a three-outcome triangle, never exceeding the contextual part's own negativity. So the negativity is not a fixed multiple of the unsavable fraction; the two measure the same thing, but in different units.

The drug example sits on the bottom rung. In example 3 the B margin differs between the two populations ($P(H_B)$ is 0.26 in one and 0.40 in the other), so the pair tables disagree on an overlap: there is no signed global law at all, and the linear gap is 0.28. Yet $\lambda^* = 0.86$: the graded explanations of section 2 are about sub-measures, and they still save 86% of the structure. In the language of contextuality-by-default [27] this is a direct influence, not contextuality; the ladder separates the two, and λ^* measures both together. Population 1 alone is on the top rung.

Level two. For two actors and three variables, theorem 13 already places every generic law on a rung. The measures follow from proposition 12: since $1 - 4w_1w_2 = m_{123}^2/\mathrm{Det}$, the weights are $\frac{1}{2}(1 \pm |m_{123}|/\sqrt{\mathrm{Det}})$, so on the $\mathbb{R}$ rung the negative weight is $\nu_2 = \frac{1}{2}(|m_{123}|/\sqrt{\mathrm{Det}} - 1)$, which tends to 0 as $\Delta_m \to 0^-$ and to ∞ as $\mathrm{Det} \to 0^+$. On the $\mathbb{C}$ rung the imaginary parts measure the failure; along the Puiseux path of section 5 the explanation for $s < 0$ is $c \pm i\sqrt{|s|}\,e$, with imaginary part $2\sqrt{|s|}\max_i |e_i|$ in d.

Proposition 17 (how a law moves between rungs; LD4–LD6)**.** *For $m = 3$ a two-type explanation moves between rungs only (a) across $\mathrm{Det} = 0$, where the weights diverge, (b) across $\Delta_m = 0$, either through a non-flat stratum or through a point of the closure that is not in the image, where one weight passes through 0 while $\max|d_i|$ diverges, or (c) across a wall of the box. In (a) and in the second case of (b) the fibre has a point at infinity: the map is not proper there.*

Proof. The rung is determined by the signs of Det and Δ_m and by the box (theorem 13). By proposition 12, $w_1w_2 = \Delta_m/\mathrm{Det}$ diverges as $\mathrm{Det} \to 0$ with $\Delta_m \neq 0$; and $\Delta_m \to 0$ with $\mathrm{Det} \to m_{123}^2 \neq 0$ sends one weight to 0 and, if only one pairwise moment vanishes, $d_k = m_{123}/((w_2 - w_1)m_{ij})$ to infinity. □

The checks confirm the closed-form rung against how each law was made (200 genuine mixtures, 200 valid laws made from signed weights, and complex cases confirmed by an exact Gröbner solve; about 8% of laws drawn at random from a Dirichlet distribution lie on the complex rung), and follow a path across all the walls (fig. 3(b)).

What the ladder says. Incoherence comes in kinds, and the kinds are the rungs. Disagreement on an overlap is the bottom rung: no number system rescues it, and the only remedy is to move the evidence (the consistency radius, the linear gap). Contextuality is the $\mathbb{R}$ rung at level one: the evidence is consistent as linear algebra but needs negative probability. At level two the same rung means actors with negative weight, and a new rung appears below it: actors that exist only as complex conjugates. The gradation of section 2 cuts across the ladder, since λ^* (or λ_2^*) measures how much of the evidence can be saved on the top rung whatever rung the whole sits on. The scheme fixes the equations; the semiring decides whether their points are probabilities. A single

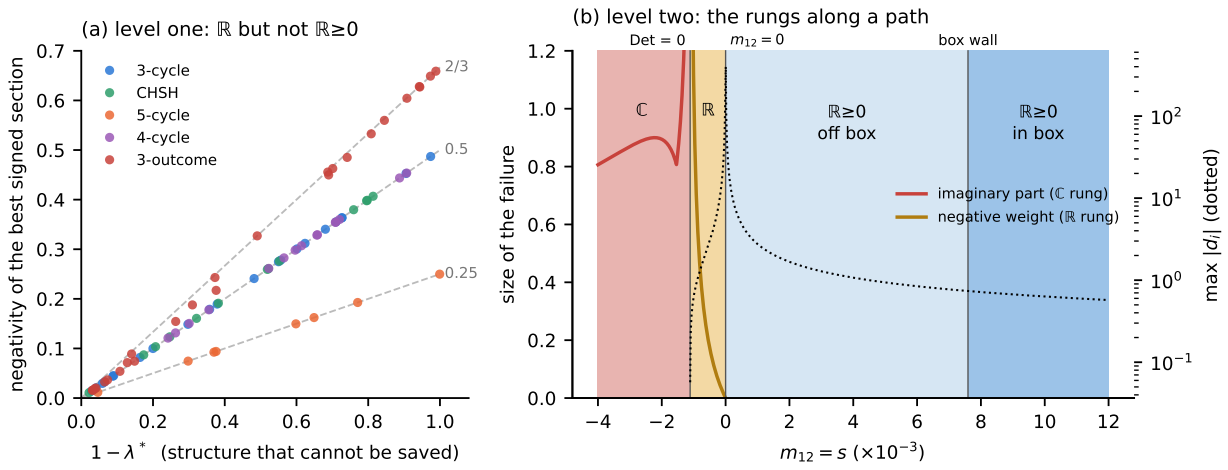


Figure 3: (a) Level one: the negativity of the best signed global section against $1 - \lambda^*$ for no-signalling families in five scenarios; each lies on or between lines whose slope is its contextual part's own negativity. (b) Level two, along the path $m_{12} = s$ ($m_{13} = m_{23} = 0.03$, $m_{123} = 0.002$, means $\frac{1}{2}$): the rung of the two-type explanation, the imaginary part on the $\mathbb{C}$ rung, the negative weight on the $\mathbb{R}$ rung, and $\max|d_i|$ (dotted), which diverges at $m_{12} = 0$. The explanation enters the box at $s \approx 0.0076$.

semiring scheme carrying both levels, with λ^* read from it, is the step this section motivates but does not take.

9 Obstructions to commutativity as structure

Everything so far is commutative: probabilities are numbers, and the order in which evidence is taken does not matter. Here we let the coefficients be matrices. A state is a density matrix ρ (Hermitian, positive semidefinite, trace 1), an observable is a projective measurement $\{A_a\}$ (orthogonal projections summing to the identity), and spatial arrangement enters through rotations: the spin-j component $n \cdot J$ along a direction n. The aim is not to discard what fails to commute but to give it a place. It finds three: the cover of contexts, the complex rung of the ladder, and the holonomy of loops. All statements are checked in `engine/check_noncomm.py` (NC1–NC8).

Commutators decide the cover. Observables that commute can be measured together. The contexts are therefore the maximal sets of pairwise commuting observables (cliques of the commutation graph), and a state ρ assigns each context its joint table $\mathrm{tr}(\rho \prod_i A_{a_i}^{(i)})$. These tables agree on overlaps automatically. If all observables commute there is one context and the tables glue (the joint spectral measure; NC1). Gluing can fail only through non-commutativity. For the CHSH observables the commutation graph has exactly the four Bell contexts, and at Tsirelson's angles [48] the engine's noncontextual fraction is $\lambda^* = 2 - \sqrt{2}$, the known value. As the angle between A_0 and A_1 opens from 0, $\|[A_0, A_1]\|$ grows from 0 and λ^* falls from 1 (NC3, fig. 4(b)). Classically the cover was given; here it is derived from the algebra, and the gluing questions of sections 2 and 8 sit on top of it.

Order is bounded by the commutator. Measuring P then Q (Lüders' rule [40]) gives $\mathrm{tr}(QP\rho PQ)$, so the order effect is $O = \mathrm{tr}\big(\rho\,(PQP - QPQ)\big)$.

Proposition 18 (NC2). *$|O| \le \|[P, Q]\|$ for every state, and the QQ equality $p(P, \bar{Q}) + p(\bar{P}, Q) = p(Q, \bar{P}) + p(\bar{Q}, P)$ holds for every pair of binary projective measurements.*

Proof. By the two-subspace theorem [37] the space splits into blocks where P and Q commute (contributing 0) and 2×2 blocks with an angle θ, where $P = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and Q projects onto $(\cos\theta, \sin\theta)$. There $PQP - QPQ = \cos^2\theta\,(P - Q)$, of norm $\cos^2\theta\,|\sin\theta| \le |\cos\theta \sin\theta| = \|[P, Q]\|$ on the block. The QQ equality follows by expanding with $\bar{P} = 1 - P$, $\bar{Q} = 1 - Q$: $P\bar{Q}P + \bar{P}Q\bar{P} = P + Q - PQ - QP$, which is symmetric in P and Q, so both sides equal $\mathrm{tr}\big(\rho(P + Q - PQ - QP)\big)$. □

The QQ equality is the test Wang, Solloway, Shiffrin and Busemeyer applied to order effects in survey questions [49]; it holds for any projective model and constrains the order effects that commutation leaves.

The canonical section lives on the complex rung. For two measurements A and B the Kirkwood–Dirac table [38, 36, 33]

$$q(a, b) = \mathrm{tr}(B_b A_a \rho)$$

has exactly the right marginals, $\sum_b q(a, b) = \mathrm{tr}(A_a \rho)$ and $\sum_a q(a, b) = \mathrm{tr}(B_b \rho)$: it is a global section of the two tables. But its entries are complex, and

$$\mathrm{Im}\, q(a, b) = \tfrac{1}{2i}\,\mathrm{tr}\big([B_b, A_a]\rho\big),$$

so the imaginary part is an expectation of commutators (NC4). If A and B commute, $B_b A_a$ is a projection and q is a law: the top rung. Real projections with a real state give a real q that can be negative: the $\mathbb{R}$ rung. Otherwise q is non-real: the $\mathbb{C}$ rung. At level one, with numbers as coefficients, the complex rung was empty (table 3); with matrices it is inhabited, and its inhabitants

are exactly the commutation obstructions. This is a finer statement than λ^*. Two contexts that do not overlap always glue classically (take the product), so $\lambda^* = 1$ there; what fails is that the section the state itself determines is not a law. Pusey showed that anomalous weak values, which are ratios of Kirkwood–Dirac entries, are proofs of contextuality [46].

Loops leave a phase, and the phase can be non-abelian. Around a loop of rank-one projections the product $\mathrm{tr}(P_1P_2P_3)$ (the Bargmann invariant [34]) is gauge-invariant. For spin $\frac{1}{2}$ its argument is half the signed solid angle of the spherical triangle, the Pancharatnam–Berry phase [44, 35]; for real projections it is real (NC5, fig. 4(d)). Complex coefficients are therefore what makes loops carry information: the phase is invisible in each probability and appears only in the order of the product. For spin j, densely sampled loops return the state with phase $j\Omega$ (NC8), so two paths to the same target with the same success probability differ by a geometric phase. With rank-two projections the loop acts on the subspace by a 2×2 unitary, and two different loops give unitaries that do not commute (NC6): the holonomy is non-abelian [50]. This is where the $\mathbb{Z}/2$ exchange of actors in section 6 goes once actors carry matrices: the discrete swap becomes one element of a continuous, non-abelian holonomy.

Probability paths. Projections that do not commute can steer. Start in $|j, j\rangle$ along n_0 and measure $n \cdot J$ along N equally spaced directions on the great circle to n_1 (angle Θ). Each step keeps the tracked outcome with probability $|d^j_{jj}(\Theta/N)|^2 = \cos^{4j}(\Theta/2N)$, so

$$P_N = \cos^{4jN}\Big(\frac{\Theta}{2N}\Big) \;\longrightarrow\; 1 \qquad (N \to \infty),$$

against $\cos^{4j}(\Theta/2)$ for the direct jump (NC7, exact for $j = \frac{1}{2}, 1, \frac{3}{2}$; fig. 4(c)). This is Zeno dragging [42, 32]. Projections that commute with each other and with the state can only condition; they never move it (NC7). Read as an analogy for molecular arrangements: a configuration that is improbable in one step can be reached with probability close to one along a path of intermediate arrangements, the route matters through the geometric phase even when the success probability does not, and the whole effect lives in the non-commutativity. Whether a given chemical system behaves this way is a physical question these toy spin models do not answer.

What survives of the scheme picture. For a fixed matrix dimension d, take the real and imaginary parts of the entries as coordinates: marginals (partial traces) are linear, Born probabilities quadratic, and $\rho \succeq 0$ a set of polynomial inequalities. Explanations are again real points of a commutative scheme, and the strata become rank conditions. Three things change. The positive cone is the positive semidefinite cone, which is not a semiring, so the ladder of section 8 becomes a ladder of cones, classical $\subset$ quantum $\subset$ signed (between local and no-signalling sits the quantum set, approximated by the NPA hierarchy [43]). Actors carrying matrices have a unitary gauge freedom, so fibres become orbits and the right object is the quotient, whose coordinate ring is generated by trace polynomials [45]. And if d is not fixed there is no single scheme: the observables generate a non-commutative algebra whose representation schemes Rep_d form a tower [39]. Whether some d explains the data is the question behind Tsirelson's problem, and it is undecidable in general [47, 41]. Our classical results are the diagonal sector of this picture. Whether the fibre over a merge point stays reduced once off-diagonal (coherence) directions are allowed is the first open scheme-theoretic question here.

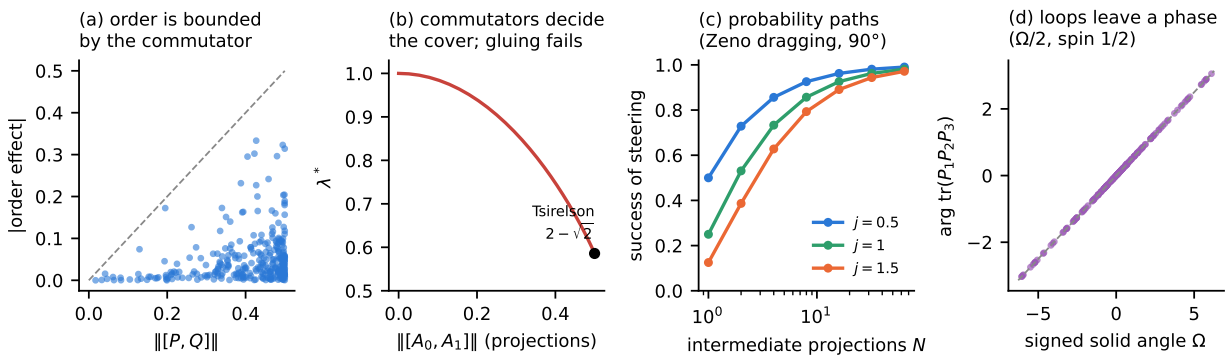


Figure 4: (a) Order effects against $\|[P,Q]\|$ for 300 random pairs of projections: all below the diagonal (proposition 18). (b) The CHSH family: as the angle between A_0 and A_1 opens, the commutator grows and λ^* falls from 1 to $2-\sqrt{2}$. (c) Zeno dragging through 90°: success against the number of intermediate projections, for spin $j = \frac{1}{2}, 1, \frac{3}{2}$. (d) The loop phase $\arg\operatorname{tr}(P_1P_2P_3)$ against the signed solid angle for 200 random spin-$\frac{1}{2}$ triangles: the line $\Omega/2$.

10 Given measurements, partly masked

Section 9 ended at a fork: either the measurements are given, or they too are unknown, in which case deciding whether some dimension explains the data is undecidable in general. Here we take the measurements as given and then hide a fraction of them, to see how the difficulty grows and how much geometry and positivity still recover. At a fixed dimension nothing becomes undecidable: the questions are first-order statements over the reals and so decidable by quantifier elimination [51]. What grows is the cost, the non-convexity, and the size of the set of explanations. All results are in `engine/check_masked.py` (MK1–MK6) and `engine/check_masked_gauge.py` (MK7).

Setting. Two qubits; Alice measures spin along $K = 5$ known directions a_i, Bob along $K = 5$ known directions b_j; every pair is a context whose 2×2 table is observed with $N = 2000$ shots, from a random entangled state mixed with 15% noise. A masked direction still has its tables observed, but the solver does not know it. With Bloch vectors s, t and correlation matrix T,

$$P(\alpha,\beta \mid a,b) = \tfrac{1}{4}\big(1 + \alpha\, s\cdot a + \beta\, t\cdot b + \alpha\beta\, a^{\top}Tb\big),$$

linear in the state for fixed directions and linear in each direction for a fixed state. With nothing masked the state is a linear least-squares fit, unique and positive within noise (MK1).

What the linear fit can see.

Proposition 19 (MK5, MK7)**.** *Let Alice's known directions span $V \subseteq \mathbb{R}^3$ and let h_A of her directions be masked; let Bob's directions span $\mathbb{R}^3$. Then the tables do not change if each hidden direction n is replaced by Mn and the state by $s \mapsto M^{-\top}s$, $T \mapsto M^{-\top}T$, for any $M \in GL_3$ that fixes V pointwise. The linear fit therefore leaves the hidden directions ambiguous along this group, cut by unit length: a family of dimension $\max\big(0,\, 3(3-\dim V) - h_A\big)$. In particular the hidden directions are located by the linear fit iff the known directions on their side span $\mathbb{R}^3$; and with three directions per side and $\dim V = 2$, the fit is the same for every candidate direction.*

Proof. Alice's data enter only through $s\cdot a$ and $a^{\top}Tb$. For a known $a \in V$, $(M^{-\top}s)\cdot a = s\cdot M^{-1}a = s\cdot a$ and likewise for T; for a hidden n, $(M^{-\top}s)\cdot Mn = s\cdot n$. The group of M fixing V pointwise

has dimension $3(3 - \dim V)$, and unit length imposes one condition per hidden direction. The dimension count is checked as the kernel of the Jacobian at the truth in seven cases ($\dim V = 3, 2, 1$; $h_A = 1, 2, 3$), all equal to the prediction (MK7). With three directions per side the nine tables carry exactly the fifteen state parameters, so for planar known directions the residual is zero at 92% of lattice points (the rest lie near the plane, where the three directions become dependent; MK5). □

When the known directions span $\mathbb{R}^3$, the linear fit has one basin: refining from the best lattice point recovers a hidden direction exactly on noiseless data (fig. 5(c)).

What the cone adds: culling impossible points. When the linear fit is blind, positivity is not. Every candidate direction implies a fitted state; if that state has a negative eigenvalue, it assigns negative probability to some measurement, and the candidate is not merely improbable but impossible. On a lattice this is one eigenvalue per point. In the planar case (two known directions in a plane and one hidden out of it, on each side) only 0.01% of the 40,000 lattice points imply a positive state; none of them is the truth, because the feasible set is a small region that a 14° lattice misses. Using the smallest eigenvalue as a score, keeping the best lattice points, and climbing from them finds the feasible set, which contains the truth, is closed under the mirror (global complex conjugation, which preserves positivity), and bounds invariants: over 1434 feasible samples the CHSH maximum [52] lies in $[2.254, 2.452]$ (truth 2.354), and a held-out correlation in $[0.159, 0.256]$ (truth 0.211) (MK6, fig. 5(d)). A mirror applied to one side only is a partial transpose, which does not preserve positivity for entangled states [53], so the cone removes one-sided ambiguities while keeping the two-sided one.

Random masking. Table 4 summarises six random trials at each level (fresh state and directions each time). Hiding one or two of ten directions always leaves at least three known on each side, so the linear fit locates them: an exhaustive lattice of 300 directions per hidden direction is culled to under 2% by the fit alone, and in every trial whose known directions are well-conditioned (9 of 12) exactly one explanation survives the cone (MK2). The exception is instructive. In one trial at 20%, with nearly coplanar known directions, the linear fit left three explanations and none of them implies a positive state, although the truth lies inside the χ^2 set: when the geometry is nearly degenerate the positive explanation sits away from the linear optima, and positivity has to drive the search, as in the planar case above, rather than be applied afterwards. At 30% and 50% the exhaustive lattice would need 300^h fits, so coordinate-wise lattice sweeps take over; they reach the best explanation found by any method (MK3). The errors are large exactly in the trials where a side keeps fewer than three independent known directions, as proposition 19 predicts.

masked	located linearly	median error	truth in 99% set	fits used	local-search success
0%	6/6	0°	6/6	1	—
10% (1 of 10)	6/6	3.0°	5/6	300	0.94
20% (2)	6/6	3.5°	6/6	90,000	0.81
30% (3)	4/6	4.4° (others 67°)	6/6	16,200	0.74
50% (5)	0/6	(all 49°)	6/6	27,000	0.69

Table 4: Masked measurements, six trials per level. "Located linearly": each side keeps known directions spanning $\mathbb{R}^3$. The error is the largest angle between a recovered and a true hidden direction. Fits: exhaustive lattice for one and two hidden directions, lattice sweeps beyond. Local search: the fraction of random starts of plain least squares that reach the best explanation.

Three further points. The coverage of the 99% χ^2 set was 11 of 12 at 10–20%, so the Gaussian approximation behind it is somewhat optimistic. The operator-free rungs of section 8 are computed from the tables alone, so masking does not touch them: the noncontextual fraction of the tables was between 0.947 and 0.967 in the sampled trials, whatever was hidden (MK4). And what can be extracted is exactly what is invariant under the residual symmetry: the CHSH maximum survives the mirror and is pinned by the cone, while a correlation with a new direction is not invariant and has a wider interval. In the language of section 9, what masked measurements leave recoverable are the functions on the quotient by the residual group, cut down by positivity.

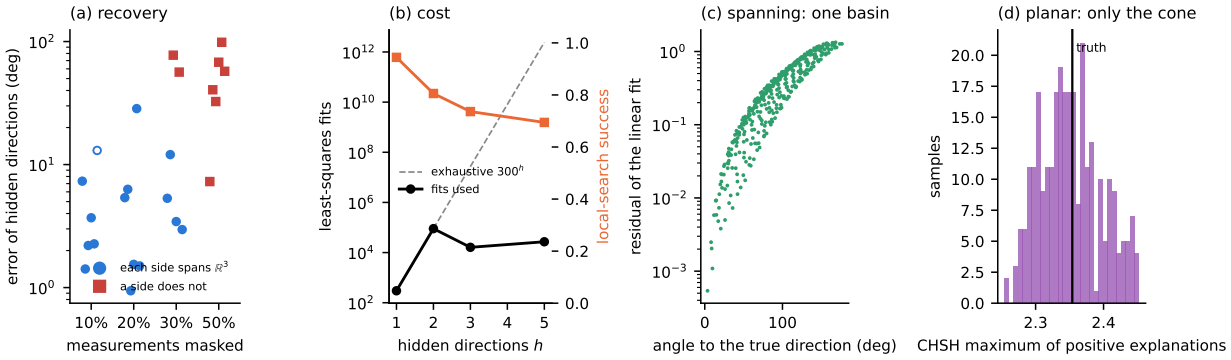


Figure 5: (a) Error of the recovered hidden directions against the masking level, split by whether each side keeps known directions spanning $\mathbb{R}^3$ (circles) or not (squares); an open marker is a trial whose truth fell outside the 99% set. (b) Cost: least-squares fits used (exhaustive lattice up to two hidden directions, sweeps beyond) against the exhaustive count 300^h, and the success rate of plain local search. (c) Spanning known directions: the residual of the linear fit over the lattice against the angle to the true direction: one basin. (d) Planar known directions: the linear fit is blind, the cone is not; the CHSH maximum over the feasible (positive) explanations against the truth.

11 Quantum graded gluing: positivity domains of hereditary quadratics

Islam, Volčič and Woerdeman study self-adjoint matrix polynomials $f = L + H^*H - K^*K$ in non-commuting variables x_j and their adjoints, evaluated on tuples of matrices of every size, and their positivity domains $\mathcal{D}(f)$, where f is positive semidefinite [54]. Graded gluing with quantum windows is exactly of this form. All statements here are checked in `engine/check_qglue.py` (QG1–QG6).

The identification. Let the windows be sets of qubits with density matrices ρ_w, and let ψ be an unnormalised global vector. The quantum form of $\alpha\, q|_w \le t_w$ is

$$\mathrm{Tr}_{\mathrm{rest}}(\psi\psi^*) \preceq \rho_w \quad \text{for every window,} \qquad \text{with mass } \alpha = \|\psi\|^2.$$

Ordering ψ as ψ_{ak} (window index a, the rest k) and putting $K = \sum_{a,k} |k\rangle\langle a|\, x_{ak}$, the constraint is $f_w(\psi) \succeq 0$ for $f_w = \rho_w^{\top} - K^*K$, a self-adjoint hereditary quadratic with constant term and no linear term. The family gives $f = \bigoplus_w f_w$, and the quantum explanations are its positivity domain. The paper's hypotheses hold (QG1): the quadratic coefficients generate the full matrix algebra

(coefficient-irreducible); the line system is minimal; the linearisation $\left(\begin{smallmatrix} I & K \\ K^* & \rho_w^\top \end{smallmatrix}\right)$ generates the full algebra, so it is an irreducible pencil and f_w is an atom; $f_w(0) = \rho_w^\top \succ 0$ exactly when ρ_w has full rank; f_w is unsignatured (positive near 0, indefinite far out); every window is active somewhere (positivity-minimal, QG2). The grading is what makes this work. The normalised problem has an equality $\mathrm{tr} = 1$, which the framework does not allow; the graded problem has inequalities only, and $\psi = 0$ is an interior point.

The walls are algebraic. By Prop. 2.21 of [54] the boundary of $\mathcal{D}_n(f)$ lies in the real free locus $\bigcup_w \{\det f_w = 0\}$, the set where some window's constraint saturates in some direction. By their Theorem 3.18, under the hypotheses above, the boundary is Zariski dense in the free locus at every level that is a multiple of some n_0. At level one we checked the local version: at the boundary point of every ray, the active window has $\det f_w = 0$ and the real gradient of $\det f_w$ is non-zero, so the boundary contains regular real points and the real locus is dense in that component (their Prop. 3.14; QG2). Figure 6(a) shows a slice. Because f has no linear term, $\psi \in \mathcal{D}$ implies $s\psi \in \mathcal{D}$ for $0 \le s \le 1$: the domain is star-shaped, and the topology changes of the kind in their Example 4.3 cannot occur in $\mathcal{D}(f)$ itself. They can only occur in its normalised slices (the quantum E_α), which we have not established here.

Theorem 20 (classical limit and dephasing bound; QG3, QG4)**.** *Let* $\lambda_Q = \max\{\mathrm{tr}\,\sigma : \sigma \succeq 0,\ \mathrm{Tr}_{\mathrm{rest}}\sigma \preceq \rho_w\ \forall w\}$ *(a semidefinite program). For every product basis,* $\lambda_Q \le \lambda^*$ *of the tables obtained by measuring every qubit in that basis. If every* ρ_w *is diagonal,* $\lambda_Q = \lambda^*$*.*

Proof. Dephasing $\mathcal{P}$ in a product basis is positive and order-preserving, and it commutes with partial traces. If σ is feasible then $\mathcal{P}(\sigma)$ is a diagonal operator with $\mathrm{Tr}_{\mathrm{rest}}\mathcal{P}(\sigma) = \mathcal{P}_w(\mathrm{Tr}_{\mathrm{rest}}\sigma) \preceq \mathcal{P}_w(\rho_w)$, so its diagonal is a classical sub-measure whose marginals are at most the tables, and its mass is $\mathrm{tr}\,\sigma$. Hence $\mathrm{tr}\,\sigma \le \lambda^*$. If the ρ_w are diagonal, every classical sub-measure is a feasible diagonal σ. □

So the quantum rung sits below every classical view of the same evidence. On random quantum families it is strictly below: 0.004 to 0.023 below the best of twenty random product bases (fig. 6(c)). For three qubits whose pairwise windows are Werner states $p\,|\Psi^-\rangle\langle\Psi^-| + (1-p)I/4$, the computed fraction is $\lambda_Q = \min\left(1, \frac{3}{2}(1-p)\right)$ at all nineteen values of p tested, equal to the classical λ^* of the Z-basis tables (fig. 6(b)). The evidence glues fully only for $p \le \frac{1}{3}$: three pairs cannot all be strongly singlet-like, which is the monogamy of entanglement seen as a gluing failure. Each value is certified by the semidefinite dual, operators $Y_w \succeq 0$ with $\sum_w Y_w \otimes I \succeq I$ and $\sum_w \mathrm{tr}(Y_w \rho_w) = \lambda_Q$ (duality gap below 10^{-9}, QG5); this dual certificate plays the role of the Farkas certificate of level one. We have the formula numerically, not as a theorem.

One pure actor is not a mixture. The paper's level one is a single pure vector ψ. Its best fraction (found by projected gradient ascent, so a lower bound) is far below the mixed fraction: at least 0.51 against 1 at $p = 0.2$, 0.40 against 0.75 at $p = 0.5$, 0.18 against 0.30 at $p = 0.8$ (fig. 6(d), QG6). At higher levels the paper's domains are free sets: closed under direct sums, unitary similarity and compression to common invariant subspaces (checked at level two, QG1). Their physical reading for us (operator-valued amplitudes on an environment of dimension n) and their relation to mixed explanations remain to be worked out.

What else the paper offers. Their Theorem 4.9 says that for unsignatured atomic hereditary quadratics with $f(0) \succ 0$, equal positivity domains (at every level) hold exactly when $f = S\tilde{f}S^*$.

For one window this says the quantum explanation set determines the window's constraint up to congruence; for a family, which is a direct sum and not an atom, it does not apply as stated. And hereditary-quadratic domains are Γ-convex, with a separation theorem (their Theorem 5.8, from [55]): a point outside is cut off by a hereditary-quadratic certificate, a size-free counterpart of the dual certificate above.

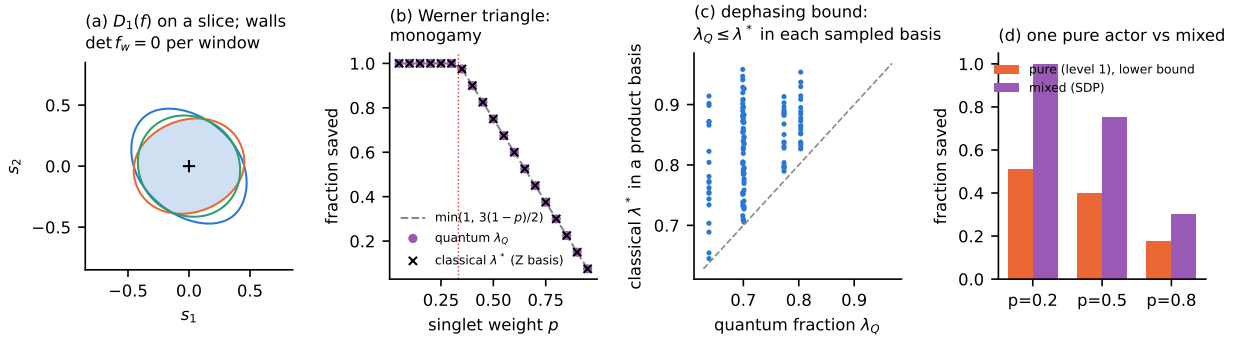


Figure 6: (a) A real two-dimensional slice of the pure domain $\mathcal{D}_1(f)$ for three random pairwise windows on three qubits: star-shaped about 0, bounded by pieces of the walls $\det f_w = 0$ (one colour per window). (b) Pairwise Werner windows: the quantum fraction, the formula $\min(1, \frac{3}{2}(1-p))$ and the Z-basis classical λ^*; full gluing only for $p \le \frac{1}{3}$. (c) The dephasing bound: for six random quantum families, λ^* in twenty random product bases against λ_Q, all above the diagonal. (d) The best pure single-actor fraction (a lower bound) against the mixed fraction for Werner windows.

12 A categorical home

The previous sections left several structures without a common setting: the cover of contexts derived from commutators, the gauge of masked measurements, the quantum rung without a fixed dimension, the information geometry, and the question of which dimension, if any, explains the data. This section places them in one framework. It does not resolve the open problems; it says where each one lives. All computations are in `engine/check_npa.py` (NP1–NP6).

Representation functors and their limits. For an algebra A of observables and a commutative ring B, the representations $\mathrm{Rep}_n(A)(B) = \mathrm{Hom}(A, M_n(B))$ form a functor represented by a commutative affine scheme, with GL_n acting by change of basis; its quotient is described by trace rings [45]. Kontsevich and Rosenberg proposed reading the geometry of A through the family $\mathrm{Rep}_n(A)$ [65], and its derived version is representation homology [57]. Li replaces M_n by an arbitrary finite-dimensional coefficient algebra M, $\mathrm{Rep}_M(A)(B) = \mathrm{Hom}(A, M \otimes B)$, and shows that non-matrix coefficients (strictly upper triangular 3×3 matrices) detect singularities of the quantum plane and the Jordan plane that every matrix coefficient misses [66]. Our fixed-dimension framing of section 9, and the linear gauge of section 10 (the subgroup of GL_3 fixing the known directions), are instances of this functor. What fails at finite dimension is made precise by the following.

Proposition 21 (holonomy against dimension; NP5)**.** *Invertible $n \times n$ matrices with $UV = qVU$ exist iff $q^n = 1$. Hence for $q = e^{2\pi ik/N}$ ($\gcd(k, N) = 1$) they exist exactly when N divides n, and for an irrational phase in no finite dimension; the corresponding algebra (the noncommutative*

torus) nevertheless has a rich category of infinite-dimensional modules [71, 58]. The Weyl relation $xy - yx = 1$ has no finite-dimensional solution at all.

Proof. Taking determinants, $\det U \det V = q^n \det V \det U$. Clock and shift matrices tensored with $I_{n/N}$ realise the relation when $N \mid n$. For the Weyl relation take traces: $0 = n$. □

So a loop that carries a phase incommensurable with every finite dimension leaves every Rep_n empty. Numerically, the best unitary residual $\|UV - qVU\|$ is 7×10^{-7} for $q^3 = 1$, $n = 3$, and 1.41, 1.04 for $n = 2, 4$, and stays above 0.75 for an irrational phase (fig. 7(c)). The physical instance is Tsirelson's problem: Slofstra built games whose perfect quantum strategies exist only in infinite dimensions [73]. This is a different phenomenon from the $\mathbb{Z}/2$ exchange of actors of section 6, which lives at a fixed dimension.

The module category, with positivity. Gabriel and Rosenberg recover a scheme from its category of quasi-coherent sheaves [61, 72], and Artin and Zhang take a category of graded modules as the definition of a noncommutative projective scheme [56]. Taking the category of modules as the space removes the dimension constraint above: infinite-dimensional modules are objects like any other. Two corrections are needed for probability. First, a plain module category has no positivity, and every result of the ladder lives in a cone, so the right category is that of $*$-representations on Hilbert spaces (or of operator systems). Second, a state is not an object but a pointed one: the GNS construction turns a state into a cyclic $*$-representation with a distinguished vector. Observables are recovered as natural endomorphisms of the forgetful functor, and the matrix polynomials of section 11 are morphisms between free modules; their free locus is the set of modules on which such a morphism fails to be invertible.

This is also what makes the framework computable. The NPA hierarchy [43, 68] builds the GNS module from words in the observables, truncated by length, and imposes positivity of its moment matrix; no dimension is fixed. It reproduces Tsirelson's bound $2\sqrt{2}$ at levels 1, $1{+}AB$ and 2 (NP1), and it gives the dimension-free quantum rung directly on tables: the largest λ with $\lambda q \le P$ cellwise for a quantum behaviour q.

Proposition 22 (the rungs on isotropic boxes; NP2, NP3)**.** *For the box with correlators $\pm v$ and unbiased marginals, $\frac{1}{2} \le v \le 1$: the classical fraction is $\lambda^* = 2(1 - v)$, and the quantum fraction is 1 for $v \le 1/\sqrt{2}$ and $(1 - v)/(1 - 1/\sqrt{2})$ beyond.*

Proof. Write the box as $\lambda q + (1-\lambda)r$ with r no-signalling. The CHSH value is linear, equal to $4v$ for the box, at most 2 for local q, at most $2\sqrt{2}$ for quantum q and at most 4 for r; so $4v \le 2\lambda + 4(1-\lambda)$, i.e. $\lambda \le 2(1-v)$, and $4v \le 2\sqrt{2}\lambda + 4(1-\lambda)$, i.e. $\lambda \le (1-v)/(1-1/\sqrt{2})$. Both bounds are attained, by local deterministic boxes and by the Tsirelson box, respectively, mixed with the PR box. □

The engine's linear program and NPA reproduce both formulas at 21 values of v to 4×10^{-10} (fig. 7(a)); on random no-signalling boxes the hierarchy is ordered, $\lambda^* \le \lambda_{\mathrm{NPA}(2)} \le \lambda_{\mathrm{NPA}(1+AB)} \le \lambda_{\mathrm{NPA}(1)} \le 1$ (fig. 7(b)). For two parties with two-outcome measurements, Jordan's lemma reduces quantum correlations to qubits, so here the dimension-free rung adds nothing beyond Rep_2; it matters where proposition 21 bites.

Contexts as a diagram, contextuality as a failed limit. De Silva and Barbosa associate to an operator algebra the diagram of Gelfand spectra of its commutative subalgebras (the spectral presheaf of Isham, Butterfield and Hamilton [63]), and extend any functor on spaces to all algebras by taking the limit over that diagram [59]. The diagram is our commutation cover. Kochen–Specker

[64] says its limit of points is empty; Gleason [62] says compatible probability assignments still exist. Contextuality sits between the two: probabilities agree across contexts but come from no measure on global points, and λ^* measures how far.

Proposition 23 (Peres–Mermin; NP6)**.** *For the nine two-qubit Pauli observables of the Peres–Mermin square [69, 67], the commutation cliques are the six lines; no ± 1 assignment respects all six line products; and for every state the six context tables agree on overlaps (a signed global section exists) while $\lambda^* = 0$.*

Proof. Five line products are $+I$ and one is $-I$, so the product of all six constraints over the nine values is $+1 = -1$. Every global assignment therefore violates some line, where it hits a cell of probability zero, so any sub-measure below the tables vanishes. The tables come from one state on jointly measurable sets, so they agree on overlaps. □

Checked for the maximally mixed state and five random states (rung $\mathbb{R}$, $\lambda^* = 0$, linear gap 0). This is state-independent contextuality, the cleanest case where points collapse but probabilities survive.

Information geometry as enrichment. Perrone enriches Markov categories (states as morphisms out of the unit, channels as morphisms) in divergences, so that composition and tensor can only contract them; data processing inequalities follow from the enrichment [70, 60]. For gluing it gives the following.

Proposition 24 (data processing for gluing; NP4)**.** *Applying local channels to the outcomes of each variable never decreases λ^*.*

Proof. Push the optimal sub-measure μ (with $\mu|_w \le t_w$) through the channels. Positivity of the channels keeps every marginal below the processed tables, and the mass is unchanged. □

Processing the evidence cannot create contextuality, consistent with the monotones of Abramsky, Barbosa and Mansfield [3]. On ten random boxes the level-$(1+AB)$ quantum fraction never decreased either (numerical, not proved). A quantum Markov category has no copy map, and that failure of copying is non-commutativity itself.

Table 5 summarises the placement. The categorical view unifies the framework; it does not by itself compute anything, which is why the NPA truncation matters, and it leaves the commutative level-two results and the open problems where they were.

earlier structure	categorical home
cover of contexts (section 9)	the diagram of commutative subalgebras (spectral presheaf)
gluing and its failure (sections 2 and 8)	limits over that diagram: points (Kochen–Specker) versus probabilities (Gleason)
the ladder of rungs	the cone in which the sections live: orthant, positive semidefinite, signed, complex
quantum rung without fixed dimension	positivity on truncated free modules (NPA)
masked directions, gauge (section 10)	the GL_n action on $\mathrm{Rep}_n(A)$ and its quotient
holonomy and phases (section 9)	relations such as $UV = qVU$; finite dimension only when $q^n = 1$
positivity domains (section 11)	morphisms of free modules; free loci as non-invertibility
information geometry	divergence-enriched Markov categories; data processing
level-two latent strata, flattening, hyperdeterminant	remain commutative algebraic geometry
Werner formula, quantum slices	open

Table 5: What the categorical view gives a home to, and what it does not.

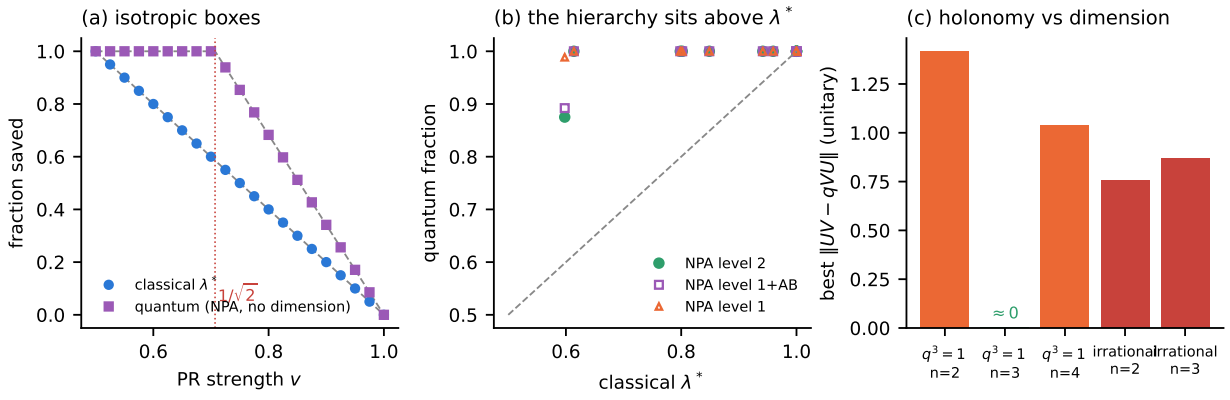


Figure 7: (a) Isotropic PR boxes: the classical fraction $\lambda^* = 2(1-v)$ and the dimension-free quantum fraction (NPA level $1+AB$), 1 up to $v = 1/\sqrt{2}$ and $(1-v)/(1-1/\sqrt{2})$ beyond (dashed: proposition 22). (b) Twelve random no-signalling boxes: the quantum fraction at three levels against λ^*. (c) The best unitary residual of $UV = qVU$ by dimension: zero only when $q^n = 1$ (proposition 21).

13 What this says about probability built on schemes

probabilistic notion	geometric form
the tables glue	$\lambda^* = 1$: the level-one fibre (a polytope) is non-empty
graded partial gluing	the slices E_α of the polytope $\{\mu \geq 0 : \mu\|_w \leq t_w\}$, $0 < \alpha \leq \lambda^*$
dissipated mass	the residues r_w of (1); $1 - \alpha$ of every table
most probable path	the entropy maximisers q_α, continuous in α
latent actors	the fibre $\mathcal{X}_P$ of the polynomial map F, an affine scheme
actors merge, or one is absent	singular strata; over an independent law, the Stanley–Reisner scheme of a simplicial complex
actors partly invisible	the non-flat locus $\{w_1 w_2 d_1 d_2 d_3 = 0\}$ $(m = 3)$
second-order obstruction ($\ell_2 \neq 0$)	$\kappa = C^\top D^2 F$; tangent cone = union of branches
abrupt change	a path of laws crossing a fold of the image; Puiseux branches $\pm\sqrt{s}$
a structure dies, a new one is born	real points of the fibre become a complex-conjugate pair (sign of the hyperdeterminant); three types fit
signed mixtures	real points with weights outside $[0, 1]$ ($\mathrm{Det} > 0 > \Delta_m$)
kinds of incoherence	rungs of the ladder $\mathbb{C} \supset \mathbb{R} \supset \mathbb{R}_{\geq 0}$: no points (direct influence), $\mathbb{R}$ not $\mathbb{R}_{\geq 0}$ (contextuality, negativity ν), $\mathbb{C}$ not $\mathbb{R}$ (level two)
commutation obstruction	the cover of contexts; order effects $\leq \|\|[P, Q]\|\|$; Kirkwood–Dirac section on the $\mathbb{C}$ rung
loops of evidence	geometric phase; non-abelian holonomy with rank-two projections
masked measurements	linear ambiguity of dimension $\max(0, 3(3 - \dim V) - h)$; the cone culls candidates implying negative probabilities; recoverable = invariants of the residual group
categorical home	contexts as a diagram of commutative subalgebras; contextuality as a failed limit; NPA truncation; $q^n = 1$ obstruction; data processing
quantum rung	positivity domain of $\bigoplus_w (\rho_w^\top - K_w^* K_w)$; walls in the free locus $\det f_w = 0$; $\lambda_Q \leq \lambda^*$ in every product basis
probability paths	Zeno dragging: success $\cos^{4jN}(\Theta/2N) \to 1$, only with non-commuting projections
dual roles	monodromy of F (a subgroup of S_k); trivial on real mixtures when $m = 3$ because the non-flat walls separate the sheets

Table 6: The dictionary.

Table 6 collects the dictionary. Level one is linear: explanations form polytopes, the gradation α slices them, and everything is a linear program. The only way the structure changes is through the combinatorics of the slices. Level two is polynomial: explanations are the real points of the fibres of F, the fibres have singular points where the actors merge or vanish, the obstruction to deforming an explanation is quadratic, and closed loops of evidence can permute the actors. The gradation lives at both levels (λ^* and λ_2^*), and so a programme for a theory of probability on schemes is: for each level, the fibre over the evidence; for each fibre, its graded slices; and for each path of evidence, the path it induces through the slices, including where it must pass through a singularity and what is lost or born there.

Related work. The noncontextual fraction and the sheaf view of local tables are due to Abramsky and coauthors [1, 3]; graded agreement in sheaves of pseudometric spaces is Robinson's consistency radius [11]; the companion report [31] develops the level-one engine. That latent-class and other

hidden-variable models are stratified, singular algebraic families is classical in algebraic statistics [8, 9], and the consequences of their singularities for estimation and testing are studied by Drton [7] and Watanabe [30]. Generic identifiability of two-type models on three or more binary variables is due to Allman, Matias and Rhodes [4]. The secant-cumulant coordinates of section 7 and the toric structure of secant varieties are due to Michałek, Oeding and Zwiernik [24]; the semialgebraic description of tensors of nonnegative rank two to Allman, Rhodes, Sturmfels and Zwiernik [12], with a book-length treatment of semialgebraic latent models by Zwiernik [26]; real rank and the hyperdeterminant to de Silva and Lim [16] and Gelfand, Kapranov and Zelevinsky [17]; Stanley–Reisner theory and flatness are standard [25, 23]. What we add is the combination: graded partial gluing at the failure of a sheaf condition, lifted to the level of hypotheses, with the singular strata, second-order obstruction, Puiseux crossing and monodromy read as what happens to latent actors when evidence moves.

Limits. The categorical section is a framework, not a theorem: its computations (NPA, the Peres–Mermin square, the $q^n = 1$ obstruction, data processing) are standard or elementary, and the monotonicity of the quantum fraction under processing is checked, not proved. The quantum section uses three qubits with pairwise windows; the Werner formula is computed and certified numerically, not proved; pure fractions are lower bounds from a non-smooth optimisation; topology changes of the normalised quantum slices were not established, because their maximisers come in continuous families. The masking study uses two qubits with random states and directions, six trials per level, and a Gaussian χ^2 calibration that proved somewhat optimistic; the positivity-only search samples the feasible set by random walks and does not certify its extent. The non-commutative section uses spin models and small random examples; its molecular reading is an analogy, not chemistry, and its scheme-theoretic part (reducedness with coherences, the Rep_d tower) is stated, not computed. The exact scheme results are for two types: the Stanley–Reisner fibre for $m = 3, 4$, the closed form and the stratification for $m = 3$, and one flat point for $m = 4$; the level-two ladder is for $m = 3$, and the level-one comparability constants are measured, not derived, beyond the upper bound. The fibres are semialgebraic (they carry the inequalities $w \geq 0$, $0 \leq a \leq 1$, $\alpha F \leq t$), so "scheme" is used for the equations and the real points with these inequalities are the explanations. All variables are binary, and the level-two results are for two types. The values of λ_2^* and the level-two intervals come from multi-start optimisation and are certified only as attained (lower bounds for λ_2^*, inner bounds for intervals); the regularity of the 4-variable loop is checked at 500 points, not proved. The examples are synthetic with exact tables: there is no sampling error, and no real trial data were used.

Next steps. (1) Certified level-two bounds by polynomial optimisation (moment and sum-of-squares relaxations), which would give upper bounds on λ_2^* to match the lower ones. (2) The images of the three-type model and their folds: where the third actor of section 5 is itself forced to merge or vanish. (3) Monodromy for $k \geq 3$ (subgroups of S_k) and its relation to the singular strata. (4) A pre-registered test on aggregated combination-trial tables, if published pair and triple tables can be found; nothing patient-level would be needed. (5) Schemes that see positivity: over $\mathbb{R}$ or $\mathbb{C}$ the linear equations of level one always have solutions when the tables agree on overlaps, so contextuality is invisible to ordinary schemes; it is a statement about $\mathbb{R}_{\geq 0}$-points. Semiring schemes (blueprints [22], tropical schemes [18]) are the natural home for gluing with positivity, with λ^* and the graded explanations as their points over the α-line. (6) The flattening stratification for $m \geq 4$ and $k \geq 3$. (7) Non-commutative level two: whether the fibre over a merge point stays reduced once coherences are allowed (qubit actors), the quantum rung by the NPA hierarchy alongside

λ^*, and whether laws on the complex rung of section 8 have qubit-actor explanations. (8) A pre-registered order-effect test (the QQ equality and the commutator bound) on aggregated survey data. (9) The quantum slices: topology of the normalised explanation sets as the evidence moves (the analogue of Example 4.3 of [54]), with continuous symmetries quotiented out; an analytic proof of the Werner formula; and the meaning of the higher levels of the free domain. (10) A scenario where the module level genuinely exceeds every fixed dimension (multi-outcome or multipartite, or a Slofstra-type game), and representation homology with non-matrix coefficients [66] as a test of where finite explanations become singular.

Reproducibility. The engine module is `engine/pce/level2.py`; `engine/check_level2.py` runs nine checks (LV1–LV9: graded explanations, the conflicting trials, the exact second derivative, the strata, the tangent cone, the Puiseux crossing, monodromy, sharpening, and level-two graded gluing) and writes `level2/level2_results.json`; `level2/fig_level2.py` draws fig. 1. `engine/check_scheme.py` runs six exact symbolic checks (SC1–SC6: the moment normal form, the Stanley–Reisner fibre and its primary decomposition, the closed-form inverse and Jacobian, the hyperdeterminant, the $m = 3$ strata, and the flat point for $m = 4$) in a few seconds and writes `level2/scheme_results.json`; `level2/fig_scheme.py` draws fig. 2. The exploratory scripts are in `level2/scheme/`. `engine/pce/ladder.py` computes the rungs and their measures; `engine/check_ladder.py` runs six checks (LD1–LD6) and writes `level2/ladder_results.json`; `level2/fig_ladder.py` draws fig. 3. `engine/pce/noncomm.py` implements spin rotations, sequential measurement, the commutation cover, the Kirkwood–Dirac section, Bargmann invariants and Zeno dragging; `engine/check_noncomm.py` runs eight checks (NC1–NC8) and writes `level2/noncomm_results.json`; `level2/fig_noncomm.py` draws fig. 4. `engine/pce/masked.py` implements the two-qubit tables, the linear and positivity-constrained fits, the lattice culls, sweeps and the feasible-region search; `engine/check_masked.py` (MK1–MK6, about 13 minutes) writes `level2/masked_results.json`, `engine/check_masked_gauge.py` (MK7) checks the dimension formula, and `level2/fig_masked.py` draws fig. 5. `engine/pce/qglue.py` builds the hereditary quadratics, their linearisations and the pure and mixed quantum fractions (semidefinite programs solved with SCS through cvxpy); `engine/check_qglue.py` (QG1–QG6) writes `level2/qglue_results.json`; `level2/fig_qglue.py` draws fig. 6. `engine/pce/npa.py` implements the NPA levels 1, $1+AB$ and 2 for bipartite binary scenarios and the dimension-free quantum fraction; `engine/check_npa.py` (NP1–NP6) writes `level2/npa_results.json`; `level2/fig_npa.py` draws fig. 7.

References

[1] S. Abramsky and A. Brandenburger, The sheaf-theoretic structure of non-locality and contextuality. *New J. Phys.* **13** (2011), 113036.

[2] S. Abramsky and A. Brandenburger, An operational interpretation of negative probabilities and no-signalling models. In *Horizons of the Mind*, Lecture Notes in Computer Science **8464**, Springer, 2014; arXiv:1401.2561.

[3] S. Abramsky, R. S. Barbosa and S. Mansfield, Contextual fraction as a measure of contextuality. *Phys. Rev. Lett.* **119** (2017), 050504.

[4] E. S. Allman, C. Matias and J. A. Rhodes, Identifiability of parameters in latent structure models with many observed variables. *Ann. Statist.* **37** (2009), 3099–3132.

[5] C. Berge, *Topological Spaces*. Oliver and Boyd, 1963.

[6] J. V. Burke and M. C. Ferris, Weak sharp minima in mathematical programming. *SIAM J. Control Optim.* **31** (1993), 1340–1359.

[7] M. Drton, Likelihood ratio tests and singularities. *Ann. Statist.* **37** (2009), 979–1012.

[8] D. Geiger, D. Heckerman, H. King and C. Meek, Stratified exponential families: graphical models and model selection. *Ann. Statist.* **29** (2001), 505–529.

[9] L. Pachter and B. Sturmfels (eds.), *Algebraic Statistics for Computational Biology*. Cambridge University Press, 2005.

[10] S. Popescu and D. Rohrlich, Quantum nonlocality as an axiom. *Found. Phys.* **24** (1994), 379–385.

[11] M. Robinson, Assignments to sheaves of pseudometric spaces. *Compositionality* **2** (2020); doi:10.32408/compositionality-2-2.

[12] E. S. Allman, J. A. Rhodes, B. Sturmfels and P. Zwiernik, Tensors of nonnegative rank two. *Linear Algebra Appl.* (2015); arXiv:1305.0539.

[13] E. Casas-Alvero, *Singularities of Plane Curves*. London Mathematical Society Lecture Note Series **276**, Cambridge University Press, 2000.

[14] J. F. Clauser, M. A. Horne, A. Shimony and R. A. Holt, Proposed experiment to test local hidden-variable theories. *Phys. Rev. Lett.* **23** (1969), 880–884.

[15] D. A. Cox, J. Little and D. O'Shea, *Ideals, Varieties, and Algorithms*, 4th ed. Springer, 2015.

[16] V. de Silva and L.-H. Lim, Tensor rank and the ill-posedness of the best low-rank approximation problem. *SIAM J. Matrix Anal. Appl.* **30** (2008), 1084–1127.

[17] I. M. Gelfand, M. M. Kapranov and A. V. Zelevinsky, *Discriminants, Resultants, and Multidimensional Determinants*. Birkhäuser, 1994.

[18] J. Giansiracusa and N. Giansiracusa, Equations of tropical varieties. *Duke Math. J.* **165** (2016), 3379–3433.

[19] M. Kuranishi, On the locally complete families of complex analytic structures. *Ann. of Math.* **75** (1962), 536–577.

[20] T. Lada and J. Stasheff, Introduction to sh Lie algebras for physicists. *Internat. J. Theoret. Phys.* **32** (1993), 1087–1103.

[21] J. M. Landsberg, *Tensors: Geometry and Applications*. Graduate Studies in Mathematics **128**, American Mathematical Society, 2012.

[22] O. Lorscheid, The geometry of blueprints. Part I. *Adv. Math.* **229** (2012), 1804–1846.

[23] H. Matsumura, *Commutative Ring Theory*. Cambridge University Press, 1986.

[24] M. Michałek, L. Oeding and P. Zwiernik, Secant cumulants and toric geometry. *Int. Math. Res. Not.* (2015); arXiv:1212.1515.

[25] E. Miller and B. Sturmfels, *Combinatorial Commutative Algebra*. Graduate Texts in Mathematics **227**, Springer, 2005.

[26] P. Zwiernik, *Semialgebraic Statistics and Latent Tree Models*. Chapman and Hall/CRC, 2016.

[27] E. N. Dzhafarov and J. V. Kujala, Context–content systems of random variables: the contextuality-by-default theory. *J. Math. Psych.* **74** (2016), 11–33.

[28] A. J. Hoffman, On approximate solutions of systems of linear inequalities. *J. Res. Nat. Bur. Standards* **49** (1952), 263–265.

[29] A. Meurer et al., SymPy: symbolic computing in Python. *PeerJ Comput. Sci.* **3** (2017), e103.

[30] S. Watanabe, *Algebraic Geometry and Statistical Learning Theory*. Cambridge University Press, 2009.

[31] V. K. Awasthi, The Posterior Coherence Engine: separating uncertainty from incoherence. Technical report, October 2026.

[32] Y. Aharonov and M. Vardi, Meaning of an individual "Feynman path". *Phys. Rev. D* **21** (1980), 2235.

[33] D. R. M. Arvidsson-Shukur et al., Properties and applications of the Kirkwood–Dirac distribution. *New J. Phys.* (2024); arXiv:2403.18899.

[34] V. Bargmann, Note on Wigner's theorem on symmetry operations. *J. Math. Phys.* **5** (1964), 862–868.

[35] M. V. Berry, Quantal phase factors accompanying adiabatic changes. *Proc. R. Soc. Lond. A* **392** (1984), 45–57.

[36] P. A. M. Dirac, On the analogy between classical and quantum mechanics. *Rev. Mod. Phys.* **17** (1945), 195–199.

[37] P. R. Halmos, Two subspaces. *Trans. Amer. Math. Soc.* **144** (1969), 381–389.

[38] J. G. Kirkwood, Quantum statistics of almost classical assemblies. *Phys. Rev.* **44** (1933), 31–37.

[39] L. Le Bruyn, *Noncommutative Geometry and Cayley-smooth Orders*. Chapman and Hall/CRC, 2008.

[40] G. Lüders, Über die Zustandsänderung durch den Meßprozess. *Ann. Phys.* **443** (1951), 322–328.

[41] Z. Ji, A. Natarajan, T. Vidick, J. Wright and H. Yuen, MIP* = RE. arXiv:2001.04383, 2020.

[42] B. Misra and E. C. G. Sudarshan, The Zeno's paradox in quantum theory. *J. Math. Phys.* **18** (1977), 756–763.

[43] M. Navascués, S. Pironio and A. Acín, Bounding the set of quantum correlations. *Phys. Rev. Lett.* **98** (2007), 010401.

[44] S. Pancharatnam, Generalized theory of interference and its applications. *Proc. Indian Acad. Sci. A* **44** (1956), 247–262.

[45] C. Procesi, The invariant theory of $n \times n$ matrices. *Adv. Math.* **19** (1976), 306–381.

[46] M. F. Pusey, Anomalous weak values are proofs of contextuality. *Phys. Rev. Lett.* **113** (2014), 200401.

[47] W. Slofstra, The set of quantum correlations is not closed. *Forum Math. Pi* **7** (2019), e1.

[48] B. S. Cirel'son, Quantum generalizations of Bell's inequality. *Lett. Math. Phys.* **4** (1980), 93–100.

[49] Z. Wang, T. Solloway, R. M. Shiffrin and J. R. Busemeyer, Context effects produced by question orders reveal quantum nature of human judgments. *Proc. Natl. Acad. Sci. USA* **111** (2014), 9431–9436.

[50] F. Wilczek and A. Zee, Appearance of gauge structure in simple dynamical systems. *Phys. Rev. Lett.* **52** (1984), 2111–2114.

[51] S. Basu, R. Pollack and M.-F. Roy, *Algorithms in Real Algebraic Geometry*, 2nd ed. Springer, 2006.

[52] R. Horodecki, P. Horodecki and M. Horodecki, Violating Bell inequality by mixed spin-$\frac{1}{2}$ states: necessary and sufficient condition. *Phys. Lett. A* **200** (1995), 340–344.

[53] A. Peres, Separability criterion for density matrices. *Phys. Rev. Lett.* **77** (1996), 1413–1415.

[54] A. Islam, J. Volčič and H. J. Woerdeman, Positivity of hereditary quadratic NC polynomials and x^*x-convex domains. arXiv:2610.14053, 2026.

[55] M. Jury, I. Klep, M. E. Mancuso, S. McCullough and J. E. Pascoe, Noncommutative partial convexity via Γ-convexity. *J. Geom. Anal.* **31** (2021), 3137–3160.

[56] M. Artin and J. J. Zhang, Noncommutative projective schemes. *Adv. Math.* **109** (1994), 228–287.

[57] Yu. Berest, G. Felder and A. Ramadoss, Derived representation schemes and noncommutative geometry. In *Expository Lectures on Representation Theory*, Contemp. Math. **607**, Amer. Math. Soc., 2014, 113–162.

[58] A. Connes, *Noncommutative Geometry*. Academic Press, 1994.

[59] N. de Silva and R. S. Barbosa, Contextuality and noncommutative geometry in quantum mechanics. *Comm. Math. Phys.* **365** (2019), 375–429.

[60] T. Fritz, A synthetic approach to Markov kernels, conditional independence and theorems on sufficient statistics. *Adv. Math.* **370** (2020), 107239.

[61] P. Gabriel, Des catégories abéliennes. *Bull. Soc. Math. France* **90** (1962), 323–448.

[62] A. M. Gleason, Measures on the closed subspaces of a Hilbert space. *J. Math. Mech.* **6** (1957), 885–893.

[63] C. J. Isham and J. Butterfield, A topos perspective on the Kochen–Specker theorem: I. Quantum states as generalized valuations. *Internat. J. Theoret. Phys.* **37** (1998), 2669–2733.

[64] S. Kochen and E. P. Specker, The problem of hidden variables in quantum mechanics. *J. Math. Mech.* **17** (1967), 59–87.

[65] M. Kontsevich and A. L. Rosenberg, Noncommutative smooth spaces. In *The Gelfand Mathematical Seminars, 1996–1999*, Birkhäuser, 2000, 85–108.

[66] G. Li, Derived representation schemes with arbitrary coefficients and associative smoothness. arXiv:2607.19654, 2026.

[67] N. D. Mermin, Simple unified form for the major no-hidden-variables theorems. *Phys. Rev. Lett.* **65** (1990), 3373–3376.

[68] M. Navascués, S. Pironio and A. Acín, A convergent hierarchy of semidefinite programs characterizing the set of quantum correlations. *New J. Phys.* **10** (2008), 073013.

[69] A. Peres, Incompatible results of quantum measurements. *Phys. Lett. A* **151** (1990), 107–108.

[70] P. Perrone, Categorical information geometry. In *Geometric Science of Information (GSI 2023)*, Lecture Notes in Computer Science, Springer, 2023.

[71] M. A. Rieffel, C*-algebras associated with irrational rotations. *Pacific J. Math.* **93** (1981), 415–429.

[72] A. L. Rosenberg, The spectrum of abelian categories and reconstruction of schemes. In *Rings, Hopf Algebras, and Brauer Groups*, Lecture Notes in Pure and Appl. Math. **197**, Dekker, 1998, 257–274.

[73] W. Slofstra, Tsirelson's problem and an embedding theorem for groups arising from non-local games. *J. Amer. Math. Soc.* **33** (2020), 1–56.